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# Multimodal Teleoperation of a Wearable Dual-Arm Supernumerary Robotic System

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## Abstract

The base of a supernumerary robotic limb is a person. This project built a teleoperation system for two seven-degree-of-freedom arms carried on a worn backpack frame and commanded by a *second* person, gave it a safety layer appropriate to a manipulator standing beside somebody who is not commanding it, and ran a first pilot comparing direct control against shared-autonomy assistance.

A body-worn master supplies position from whichever sensor observes each component most directly, and classifies all fourteen channels so a failing sensor degrades the system announcedly rather than silently. Six of fourteen met the health criterion; failing ones injected 54.5 mm of unrequested motion. Vertical and fore-and-aft command track correctly; lateral does not, and no axis mapping fixes it. The two arms share no reachable point under the as-built mount; a costed mount change raised home clearance from 0.1610 m to 0.2202 m. The task set solves 13 927 inverse-kinematics calls with no failures; replacing the motion generator removed 51.3 mm of unrequested excursion; and perception on four physical cameras fits a work surface to 1.43 mm RMS.

Under ethical approval (SETREC 7111986), four operators completed sixteen sessions comparing direct VR control against shared-autonomy assistance: assisted sessions averaged shorter completion times and fewer re-grips, with unchanged hand travel and sub-5 mm lag. The pilot has no separate wearer and is pilot-scale; nothing was validated on a loaded physical arm, and the two-person study this system was built for remains the target for the work that follows.

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# Chapter 1

## Introduction

A supernumerary robotic limb is an extra arm worn on the body, working alongside the wearer’s own limbs rather than replacing or rehabilitating them [YHC<sup>+</sup>21, P<sup>+</sup>21]. What separates such a device from a fixed manipulator is consequential for control: the base of the robot is a person. It accelerates, sways and steps, and it cannot be told to hold still.

This thesis reports work on the MUlti-limb Virtual Environment, known within Imperial College London as the Dr. Octopus platform [Imp], which mounts two Kinova Gen3 seven-degree-of-freedom arms [Kin22] on a backpack frame and commands them from a *second* person: an operator who cannot feel the arms, controlling them for a wearer who cannot command them. No prior work has measured both what shared autonomy buys such an operator and what it costs the person carrying the arms, in one system, under the same trial (Chapter 2); that gap is what this project is built to close.

Two premises, argued and tested throughout, shaped the engineering. First, commanding fourteen joints through a body-worn master with no force feedback is a usability problem before it is a control problem, so position is taken from whichever sensor observes each component best rather than from the slave’s full kinematic chain. Second, a body-worn master *will* lose sensing channels over its life, so the system treats channel loss as an operating condition: every channel carries a health verdict, and when too many are lost the system reports position unavailable rather than fabricating one, because on a manipulator beside a person’s head a fabricated position is worse than none.

### 1.1 Objectives

1. Design and build an instrumented mannequin master reproducing the Gen3 morphology at reduced scale, characterised against recorded data.
2. Derive a master-to-slave command mapping that degrades predictably as sensing channels are lost, and quantify what each loss costs.
3. Implement a safety architecture appropriate to a manipulator mounted beside a person, validated by fault injection rather than inspection.
4. Characterise the reachable workspace of the mounted configuration and determine which manipulation tasks it can support.
5. Establish a verification methodology producing traceable evidence for every claim, and apply it throughout.

6. Run a pilot comparison of direct control against shared-autonomy assistance with real operators, and identify what stands between it and the full two-person study the architecture was designed for.

The planning report framed this project around a haptic master with physical springs and virtual slave-side impedance. That objective was not delivered: [Chapter 5](#) explains why, and [Chapter B](#) records the route in full.

## 1.2 Contributions

Each contribution below is stated with its evidence class, because the distinction between implemented, verified in simulation, measured on hardware, and measured on a pilot session is load-bearing throughout this report. A graceful-degradation architecture for a body-worn master, measured against 20 430 recorded frames; a reduced-degree-of-freedom command mapping that takes direction from gravity and magnitude from the best-conditioned joints; a safety architecture in which every blocking mechanism is observable, with fourteen of fourteen injected faults handled; a workspace characterisation showing the two arms share no reachable point under the as-built mount, together with a quantified and buildable mount correction; a jerk-limited, phase-synchronised motion generator that removes 51.3 mm of unrequested Cartesian excursion; a calibrate-then-plan perception stage that recovers six objects to within 1 mm; a stage-by-stage instrumentation of the perception pipeline exercised on four physical cameras; and a first pilot of the direct-versus-shared-autonomy comparison with four operators across three tasks, run under ethical approval once VR controllers supplied a working alternative to the master's incoherent channels. **No contribution below the pilot was validated on physical hardware carrying a load.**

The single most transferable outcome does not appear in that list. Twenty-six times across this project, a result that looked like a system failure was a defect in the measuring tool rather than in the system, and validating an instrument against a known answer before trusting a surprising reading it produces is the practice applied to every number in this report ([Section 5.4](#), [Chapter G](#)).

## 1.3 Report structure

[Chapter 3](#) summarises what was designed, built and run, including the pilot protocol and its ethical approval; the full engineering detail – the master's electronics and firmware, the system architecture, the perception and motion pipelines – is in [Chapters A](#) and [C](#) to [E](#). [Chapter 4](#) reports the headline measurements and the pilot's results; [Chapters F](#), [H](#) and [I](#) carry the full simulation campaign, the real-camera vision session and the workspace characterisation. [Chapter 5](#) interprets the results and states what they are worth; [Chapter N](#) is the extended version. [Chapter 6](#) closes against the objectives above. [Chapters J](#) and [K](#) are the protocol for the full two-person study and the validated analysis that would evaluate it, and [Chapter L](#) is the pilot session in full.

# Chapter 2

## Background

Five bodies of work bear on the system built here.

**Supernumerary robotic limbs.** Parietti and Asada’s founding demonstrations brace a limb against the environment to stabilise the wearer’s posture [PA14, PA13], which turns the wearer’s body from a disturbance source into part of a closed chain. Later reviews map the design space along consistent axes – mounting location trades reach against the moment arm imposed on the wearer, actuation trades bandwidth against worn mass – and note that the field reports embodiment far more often than task performance [YHC<sup>+</sup>21, P<sup>+</sup>21, Z<sup>+</sup>26]. Worn mass binds hardest: each Gen3 masses 8.2 kg, so the pair with harness exceeds 17 kg, and simulated wearer loads suggest the biomechanical cost is not negligible [MK24].

**The operator and the wearer as different people.** Fusion mounts a robotic head and two arms on a worn frame, driven by a remote operator [SSM<sup>+</sup>18] – but its three modes are levels of *physical coupling*, not machine autonomy, and the publication reports no user study and no performance numbers. Work on near-body interaction found that raising *apparent* autonomy (a concealed human operator, a wearer with no task of their own) raised the wearer’s physiological arousal and lowered self-reported trust [ZGD<sup>+</sup>26] – evidence that the wearer’s experience is not a free variable, though not evidence about real autonomy or an occupied wearer.

**Master interfaces for dissimilar kinematics.** Guggenheim and Asada argue for miniature physical masters on the grounds that geometric similarity supplies feedback inherently [GA21]; the argument covers similarity of *form*, not *scale*, and this system’s master-to-slave reach ratio is roughly one to three. Joint-space correspondence preserves posture but needs every master joint measured accurately; task-space correspondence decouples the chains at the cost of the slave’s redundant degree of freedom.

**Shared autonomy.** Dragan and Srinivasa formalise arbitration as policy blending, weighted by prediction confidence [DS13]; Javdani et al. recast it as a POMDP over the user’s goal and show that assistance which does not wait for high confidence helps operators finish faster [JSB15]. Neither was developed for a robot mounted on a second person, where the cost of assisting toward the wrong goal is not a slower trial but a manipulator moving toward someone who did not command it.

**Measuring embodiment, workload and trust.** Ownership over an artificial limb is measurable and decomposes into ownership, agency and location, which can move independently [BC98, LSK<sup>+</sup>08]. Workload separates into the NASA Task Load Index’s six subscales [HS88]; physical exertion is better tracked with a Borg scale [Bor82], appropriate for an uncompensated master arm or a carried mass.

## 2.1 The gap

No study measures the operator and the wearer together, under the same autonomy, in the same trial: shared autonomy has been shown to help an operator where the robot is attached to nobody, and raising apparent autonomy has been shown to lower a wearer's trust where the autonomy was a concealed human. Until both are measured in one system, the exchange rate between what shared autonomy buys the operator and what it costs the person carrying the arms is unknown. [Chapter 1](#) states the aim this gap sets; the full review, with every citation's scope stated precisely, is in [Chapter M](#).

# Chapter 3

## Methods

This chapter summarises what was designed, built and run. Full engineering detail – the master’s electronics and firmware, the complete system architecture, the perception and motion pipelines – is carried in [Chapters A and C](#) to [E](#) so that it can be checked and reproduced without lengthening the argument here.

### 3.1 The master arm

The master is a miniature mannequin reproducing the Gen3’s joint sequence at roughly a third of its scale ([Fig. A.1, Chapter A](#)), moved with the fingers of one or both hands. A Teensy 4.1 reads fourteen rotary potentiometers, two inertial measurement units, two force-sensitive resistors and two buttons, and emits one text frame at 100 Hz ([Table A.1](#)). Potentiometer counts pass through an integer exponential moving average ([Eq. \(A.1\)](#)) before conversion to degrees; the inertial sensors are read as a single burst so that all six axes come from one sampling instant.

Each channel carries a health verdict computed from its own recent history: *dead*, *incoherent* (present but not tracing a trajectory), *intermittent*, or *alive*. Validation is scoped to the channels the active mode actually consumes, so a dead channel the mode does not read cannot reject a frame, and when no well-conditioned channel remains the system reports position *unavailable* rather than extrapolating one.

### 3.2 From master pose to arm command

The commanded position is built in spherical form about the master’s shoulder rather than by forward kinematics from seven joint angles ([Eqs. \(C.1\) to \(C.3\)](#)): elevation from the gravity vector, azimuth from the shoulder roll, and radius from the forward-kinematic magnitude of the bend joints. Each component is taken from whichever sensor observes it most directly, so a dead roll channel can be passed through as zero without corrupting the radius. The commanded end-effector position is an anchor plus a scaled displacement from a reference latched at clutch engagement ([Eq. \(C.4\)](#)), so releasing and re-engaging the clutch indexes like a mouse lifted and set down again, and the reachable set is bounded by the robot rather than by the master’s own range.

Five modes are delivered ([Table 3.1](#)); one safety stack serves all of them, and dropping any shared mechanism is refused regardless of mode. The desk arrangement the laboratory actually operates has the operator seated across the room from the wearer, holding the VR controllers as a six-degree-of-freedom input device and watching the physical robot directly rather than wearing the headset ([Section C.3.1](#)); the alignment between the operator’s frame and the robot’s is a measured yaw rather than a mirror, because facing someone and copying them is a reflection that would invert every commanded orientation while positions looked correct.

Table 3.1: Operating modes, condensed from Table C.1. Autonomy runs at a quarter of teleoperation’s speed by design: nobody is holding a master, so the only bound on a wrong motion is how long it takes to happen.

| Mode                  | Input                         | Robot decides                  | $v_{\max}$ |
|-----------------------|-------------------------------|--------------------------------|------------|
| 01/02 direct          | mannequin or VR controllers   | nothing                        | 0.60 rad/s |
| 03/04 shared autonomy | master or controllers, vision | wrist orientation              | 0.60 rad/s |
| 06 full autonomy      | typed or spoken goal          | target, approach, grasp, place | 0.15 rad/s |

### 3.2.1 Safety architecture

Collision avoidance is graduated in three stages (Section C.2.1): collision-aware inverse kinematics, redundancy re-seeding on rejection, and a geometric clearance floor of 150 mm measured from the transform tree between any moving arm link and any wearer primitive – not from the semantic collision description, which excludes the proximal pairs a shoulder mount actually threatens. Every mechanism able to stop the arm registers by name, must declare how it clears, and expires into an explicit *unknown* state (still treated as blocking) if the loop asserting it stops running, because a blocker that silently keeps its last state is indistinguishable from one that is working. Three independent triggers latch the emergency stop until reset: an explicit call, both master buttons held together, and a dead-man keyed on the *source* timestamp of the master data, not on message arrival – the correction that makes it fire on the failure mode that actually occurs, since a degraded master keeps publishing stale values at full rate rather than falling silent.

## 3.3 Perception, planning and motion

A calibrate-then-plan stage (Chapter D) replaced declared object and surface coordinates with measured ones: a wrist-camera sweep that holds one attitude per pass and translates, printer-bed-probe style, fits the working surface and segments objects from the image rather than clustering a point cloud, and hands the result to an RRT-Connect planner [KL00] evaluated against the same moving-chain clearance the safety layer enforces. Detection is open-vocabulary (YOLO-World [CSG<sup>+</sup>24] with FastSAM [ZDA<sup>+</sup>23]), falling back to a hue-and-depth filter, and grasp synthesis is the geometric principal-axis method of Eqs. (D.1) and (D.2) rather than a learned one, chosen because it can *name* the reason for a refusal and fits the 4 GiB of video memory available on this machine.

The per-joint step limiter that previously constituted the entire motion generator for teleoperation left the commanded hand 51.3 mm off the path its own inverse-kinematics solution implied (Section E.1); it was replaced with Ruckig [BK21], a jerk-limited, phase-synchronised generator in which every joint is driven from a single scalar phase (Eq. (E.1)), so the commanded configuration is on the line between endpoints by construction. Replaying recorded joint commands onto the physical arm shows every joint parks  $0.305^\circ$  short of its target on the side it approached from (Section E.3); the two candidate models – a terminal offset and a proportional gain error – are distinguishable only at short travel, and propagating the error through the manipulator Jacobian selects the terminal offset at  $r = 0.923$ .

## 3.4 Verification methodology

Twenty-six times in this project a result that looked like a system failure was a defect in the measuring tool (Chapter G). The practice that follows from it is applied throughout: every analysis script carries a known-answer test, synthetic data is used only where its ground truth is *constructed* rather than rendered, and a result

contradicting an earlier measurement is treated as an instrument check until the two are reconciled. Because the inverse-kinematics solver restarts randomly, every reachability claim in this report is ten repeats over a path densified to 20 mm, not single calls at the endpoints, and a pose that is not solid on all ten repeats is not reported as reachable.

### 3.5 Ethics and the pilot study

**This study was approved by Imperial College London’s Science, Engineering and Technology Research Ethics Committee, approval number 7111986.** All participants gave informed consent before taking part. Participant data are anonymised throughout this report as P1–P4; `write_manifest()` refuses to record a name, email address, date of birth, address or telephone number, so anonymity is enforced in the code that writes the session record rather than only in this text.

The full two-person protocol ([Chapter J](#)) crosses direct teleoperation against shared autonomy over five tasks with a separate wearer, and was designed in full before this report but could not be run: it depends on master channels that were incoherent ([Chapter H](#)), and on an approved protocol that did not yet exist. Both blockers have since partly cleared. Approval was obtained, and VR controllers – already a delivered input mode ([Table 3.1](#)) – supply a six-degree-of-freedom command that does not depend on the master’s damaged channels, which made a first, smaller pilot practical ahead of a full session with a separate wearer.

Four operators (P1–P4) each completed between two and four recorded sessions, performing one of three manipulation-style tasks – object tracking, target reaching, and position matching – once under direct VR control and once with shared-autonomy assistance active, using the desk arrangement of [Section C.3.1](#). Coverage across participant, task and condition is uneven, which [Section 4.4](#) reports rather than hides. Every sample was logged automatically at 20 Hz – controller and end-effector pose, clutch engagement, and the VR link’s own reported spatial lag – together with an event log of every clutch and emergency-stop transition, giving completion time, hand-travel distance, re-grip count, the active-control fraction of the session, and mean tracking lag per session without any manual timing. This pilot measures *operator* performance only; it has no separate wearer and therefore does not address the wearer-side hypotheses (H3, H4) of the full protocol, which remain for the session [Chapter J](#) specifies.

# Chapter 4

## Results

Every number below is labelled with the strength of evidence behind it, and **no result before the pilot in [Section 4.4](#) was validated on a physical Kinova arm carrying a load**: the great majority of this project is simulated against mock hardware, using the same kinematic description, inverse-kinematics solver and safety code as the delivered system, plus recorded data from the physical master and four physical cameras. [Chapter H](#) maps every number below to its full derivation.

### 4.1 The command mapping and its safety layer

Channel health, computed over *distinct* sensor updates rather than transmitted rows (roughly 82 to 88 % of consecutive rows repeat, which had previously been misread as dead channels), found six of fourteen channels alive. Replaying the recorded incoherent traces with the operator’s input held constant shows those channels alone inject 54.5 mm of commanded motion on the left arm and 0.014 mm on the right – motion nobody asked for, on a master whose whole reach is 272 mm. Regressing true reach on the surviving channels shows the left arm retains no observable radial dimension at all ( $R^2 = 0.133$ ) while the right retains it almost completely ( $R^2 = 0.986$ ), which names the repair precisely as two channels on the left arm and predicts that repairing any other channel would not restore reach.

Vertical and fore-aft command axes track correctly on both arms. The lateral axis does not: two master motions intended  $90^\circ$  apart arrive  $137.7^\circ$  apart at the commanded end effector, and because a rotation preserves angles, no axis remapping or recalibration can turn a non-orthogonal measured triad into an orthogonal one. The cause is mechanical – the first joint’s axis does not tilt as the arm raises, so its effect on the tip scales with how bent the arm already is – and is recorded as a geometric limitation of the sensing choice rather than a calibration defect.

The safety layer handled fourteen of fourteen injected faults (channel dropout, board disconnect and reconnect, emergency stop mid-motion, wrong-object latching, stale detections, session loss, clock non-monotonicity and more). Two needed several attempts, and both were the same class of error: a mechanism keyed on the wrong observable. The dead-man did not trip on ten seconds of total master silence because the master substitutes stale data and keeps publishing at full rate; keying the dead-man on the data’s own source timestamp fixed it. An earlier version of the same emergency-stop path also called a blocking real-driver shutdown first, adding a measured 4 s between the decision to stop and the arm actually stopping; the fix parks the arm first, from local publishes only.

## 4.2 Workspace and motion

Over a 63-cell grid spanning the frontal working volume with both arms verified at the home pose, **no cell was reachable by both arms** simultaneously, and the result survives freeing the wrist yaw and adding solver retries. A parametric mount sweep shows this is not intrinsic: moving the mounts inboard, forward and down with additional forward tilt raises shared cells to 22 of 63 within a buildable 0.20 m separation. That mount change was subsequently applied, raising clearance at the home pose from 0.1610 m (a reading of the mount itself, since no joint moves it) to 0.2202 m, and moving the pick-and-place placement targets from 0.530 m off centre to 0.290 m. Neither arm crosses the body centre line at any surface height or distance in 3360 swept configurations; the constraint is the wearer’s torso, not their arm posture, since deleting the wearer’s arms from the model entirely moves the boundary by at most 75 mm.

The per-joint step limiter that constituted the whole of teleoperation motion generation left the hand 51.3 mm (left arm) and 39.2 mm (right) off the straight line its own inverse-kinematics solution implied, against a 30 mm grasp capture gate, and commanded peak joint speeds 12.53 times the manufacturer’s declared limit because it never read the limits file. Sweeping its step size shows the excursion does not tend to zero with a smaller step; it settles at 68 mm, which is the structural signature of an algorithm that cannot stay on the line rather than a tuning problem. The jerk-limited, phase-synchronised replacement is 0.0 mm off the line and at 1.00 times the joint-speed limit.

Replaying recorded joint commands on the physical arm shows every joint parks  $0.305^\circ$  short of its target, on the side it approached from – a terminal offset rather than noise, confirmed by propagating the error through the manipulator Jacobian ( $r = 0.923$ ) rather than by a proportional-gain model. A single scalar per arm, applied as overshoot in the direction of travel, removes 74 to 76 % of it. Folding this correction into the positioning error budget, together with a forward-kinematically derived (rather than asserted) pad offset and a measured-surface correction to the object-centring bias, takes the combined budget from 18.64 to 12.91 mm in quadrature and from 33.39 to 19.99 mm worst case – inside the 30 mm requirement for the first time, *provided* the still-unmeasured camera-to-wrist transform is small. The first sustained attempt to drive a physical arm from its own wrist camera measured that transform directly and found it is not: four poses of a stationary table disagreed by  $8.45^\circ$ , which is 34 to 59 mm of position error over the working range ([Section H.7](#)).

## 4.3 Perception and the task set

Instrumented at thirty-three stages per camera and pointed at four physical cameras for the first time ([Chapter F](#)), the wrist cameras fit a support plane at 1.43 mm RMS with 74 % inlier support at working distance – within a factor of three of the same code against a renderer, and the first evidence that the surface-fitting stage transfers to hardware. No grasp closed on any camera: the narrowest axis of the nearest object measured 144 mm against an 85 mm jaw, a statement about the scene rather than the planner. The same session found, within its first hour, that three files independently asserted the scene camera’s mounting orientation and none had measured it; the disagreement had silently inverted every earlier scene figure, and it was invisible to every geometric metric because a consistent rotation preserves every distance – it was found only by comparing one camera’s picture against another’s.

Within simulation, the full task set solves 13 927 inverse-kinematics calls with zero failures and zero breaches of the 150 mm wearer-clearance floor, and reproduces field for field on re-run. Sweeping the intent estimator that drives shared autonomy against a mis-declared table shows why the perception stage matters operationally and not only geometrically: working from declared coordinates, the estimator becomes *confidently wrong* 50 % of

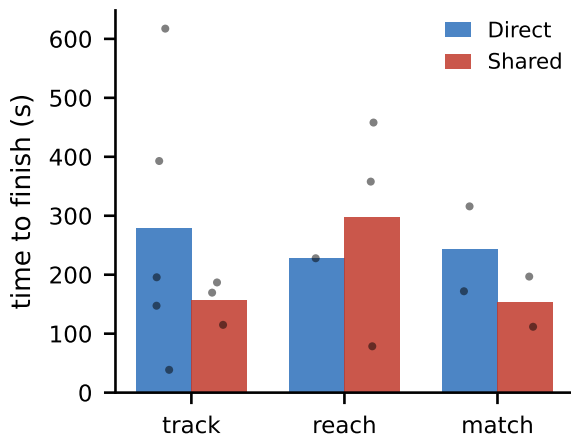
the time once the declaration is one object-pitch out, while working from measured coordinates it stays at 0.0 % confidently wrong across the same sweep.

## 4.4 A pilot of direct control against shared autonomy

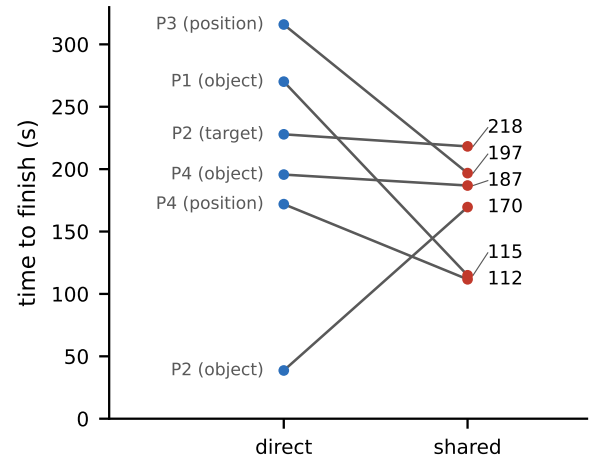
Four operators (P1–P4) completed sixteen recorded sessions under the protocol of [Section 3.5](#): direct VR control and shared-autonomy-assisted control, each on one of three tasks (object tracking, target reaching, position matching). Coverage is uneven by design – not every operator performed every task under both conditions in the time available – so the pooled comparison below mixes tasks together and the paired comparison that follows it is the part of the data that isolates the condition cleanly. Full session-by-session data, the coverage table, and the supplementary figures are in [Chapter L](#).

Table 4.1: Sixteen pilot sessions, pooled by condition. Duration includes any pausing or hesitation and is a session-length rather than a pure motor-execution measure. Every e-stop hold in this pilot occurred once, near the end of the session, to stop the recording, and is excluded as a procedural marker rather than a safety event.

| Measure ( $n = 8$ sessions per condition)         | Direct | Shared autonomy |
|---------------------------------------------------|--------|-----------------|
| Time to finish, mean (s)                          | 263    | 209             |
| Hand travel distance, mean (m)                    | 2.55   | 2.55            |
| Re-grips per session, mean                        | 29.4   | 23.9            |
| Share of session actively steering, mean          | 58 %   | 64 %            |
| Robot’s lag behind the operator’s hand, mean (mm) | 2.8    | 3.0             |



(a) Pooled across all sixteen sessions. Each dot is one recorded session; the bar is the mean for that task and condition.



(b) The six cases where one operator performed the identical task under both conditions – the fairest comparison in the pilot, since it removes both between-operator and between-task variance.

Figure 4.1: Time to finish, pooled by task (a) and paired within operator and task (b). Session length includes setup and hesitation, so this is the least precise of the pilot’s measures; the supplementary measures in [Chapter L](#) are drawn from the position trace directly and are not subject to the same confound.

Pooled across all sixteen sessions ([Table 4.1](#), [Fig. 4.1a](#)), shared autonomy sessions averaged shorter completion times (209 s against 263 s) and fewer re-grips per session (23.9 against 29.4), with essentially unchanged hand-travel distance and a robot lag under 3.1 mm in both conditions. [Figure 4.1b](#) isolates the six cases in which

the *same* operator performed the *same* task under both conditions: five of six show a shorter time under shared autonomy, and the median improvement is consistent with the pooled figure.

This pilot measures operator performance only, at pilot scale (four operators, uneven task coverage), and its e-stop, path-length and lag measures show no evidence of a safety or responsiveness cost to shared autonomy over the sessions recorded. It does not include a separate wearer and therefore cannot speak to the wearer-side hypotheses – trust and embodiment under autonomy – that motivated the two-person design of [Chapter J](#); [Section 5.5](#) treats what would be needed to extend it into that full comparison. The physical arms were polled throughout every pilot session, which lets the best-tracking session check the sim-to-real bridge live rather than by offline replay: the real arm’s own encoders follow the simulated joint they were commanded through to 1.06° RMS ([Chapter L](#)) – evidence the bridge was live and tracking, not evidence of a loaded grasp on hardware. A further three operators (M1–M3) attempted the same protocol from the mannequin master rather than VR: two track within 3°, and the third’s three sessions track 15.9 to 21.2°, an order of magnitude worse and consistent with this project’s channel-health finding that the master’s usability depends on which arm and channels a session happens to exercise ([Chapter L](#)).

# Chapter 5

## Discussion

### 5.1 What the architecture bought, and what it did not

The degradation architecture kept the system running and kept its failures visible: with eight of fourteen channels failing, the right arm retains full position sensing and the left retains direction, and the incoherent verdict exists precisely because a failing potentiometer does not go quiet – range alone calls a wide-spanning nonsense signal healthy. What it did not buy is capability. Freezing the left arm’s bend joints collapses its commanded set from a shell to a surface, and no channel-health scheme restores a measurement that was never taken. The reduced-degree-of-freedom premise succeeded exactly where a single sensor observes the quantity directly (elevation from gravity, reach from a forward-kinematic magnitude) and failed where it does not: the spherical model assumes azimuth and elevation are independent, and they are not, because the first joint’s axis does not tilt as the arm bends. That is a design rule for a successor – decompose along the axes the sensors actually observe – rather than a defect to patch with a better calibration.

### 5.2 The workspace result, and what it costs

The disjoint-workspace finding is severe because it is geometric rather than a matter of control: no planner or orientation policy can produce a point neither arm can reach, so inter-arm handover was excluded from the task set before it was excluded by anything else. The cause is not the mount’s rotation but the arms’ asymmetric home joint angles, which are ground truth because the simulation-to-hardware bridge replays them onto the real arm; fixing it needed a mechanical mount change rather than a software one, and the change was costed, applied, and re-verified as a single deliberate pass because every clearance and task coordinate in the report depends on it.

### 5.3 The safety layer is thorough about distance and silent about time

The clearance floor is a *distance*, and the wearer can move. Combining the measured wearer-tracking latency with literature limb-speed ranges gives the arithmetic in [Table 5.1](#): in most modelled limb motions the limb travels further than the entire 150 mm margin before the perception chain is even aware anything has changed. This is not a defect in the implementation; it is the consequence of a fixed margin and a perception chain of roughly half a second meeting a person who can move a metre in that time, and it is the principal open safety question this report identifies rather than resolves.

Table 5.1: Limb travel during the measured wearer-tracking latency against the 150 mm clearance floor; full table and six limb-motion classes in [Chapter E](#).

| Latency        | Deliberate reach  | Quick reach or startle        |
|----------------|-------------------|-------------------------------|
| 62 ms (median) | 62 % of the floor | exceeds the floor             |
| 562 ms (worst) | exceeds the floor | exceeds it several times over |

## 5.4 Validity of the measurements

Twenty-six times in this project a measurement was wrong and the system was not ([Chapter G](#)), which is a high enough base rate to condition how every other number in this report should be read. Fifteen of the twenty-six were caught by an independently known quantity contradicting the reading rather than by inspection, which is the argument for building a cross-check into every instrument rather than trusting it harder. Every result above except the pilot and the channel measurements comes from simulation with mock hardware, whose force channels are identically zero, so any quantity derived from force is structurally undefined rather than merely unmeasured; and every latency reported is control-path latency between software stages, excluding the operator, the display and the physical arm, so it is a floor rather than an estimate.

## 5.5 What the pilot shows, and its limits

The pilot is the first evidence from real operators on this platform, and it points the same direction as the shared-autonomy literature developed for robots not attached to anybody [[DS13](#), [JSB15](#)]: shorter completion times and fewer re-grips under assistance, with no measurable cost in hand-travel distance or responsiveness. It should not be read as more than that. Four operators with uneven task coverage is a pilot-scale sample, not a powered comparison, and the pooled figures mix three tasks together; the paired comparison is the part of the pilot that isolates the condition, and it rests on six cases. Session duration includes setup and hesitation rather than pure execution time, which is why hand travel and re-grip count – both drawn from the full-resolution position trace rather than from a stopwatch – are read as the more reliable measures here. Most importantly, the pilot has no separate wearer: it measures what the operator experienced, not what carrying 17 kg of assisted motion costs a second person, which is the comparison the two-person architecture was built to make and the reason [Chapter J](#) remains the target rather than a superseded document.

## 5.6 Against the planning report

The planning report specified physical springs at the master and virtual impedance at the slave. Neither survives contact with the measured system: the arm’s cyclic control interface is unusable over the available network path, so commands go through a high-level interface at 18.4 to 18.7 Hz, and an impedance law at that rate is not an impedance law. The compliance the plan sought would need to come from the mechanism – series elastic actuation at the joint [[TGT<sup>+</sup>22](#), [LKK19](#)] – or a current-based formulation that does not require a fast torque loop [[dWKB24](#)], both hardware changes rather than software ones. The safety architecture, the degradation architecture, the workspace characterisation and the pilot were not in the plan; each followed directly from a measurement the plan did not anticipate.

# Chapter 6

## Conclusion

This project built a teleoperation system for two seven-degree-of-freedom arms mounted on a worn backpack frame, commanded by a second person through an instrumented mannequin master or VR controllers, with a safety architecture appropriate to a manipulator standing beside somebody who is not commanding it and a shared-autonomy layer above it – and ran a first pilot comparison of direct and assisted control with real operators.

### 6.1 Against the objectives

A master was built and characterised: six of fourteen channels alive, the repair identified to two specific joints. The command mapping degrades predictably – vertical and fore-aft track correctly, lateral does not, and no recalibration fixes it. The safety architecture handled fourteen of fourteen injected faults with every blocker observable by name. The workspace characterisation found the two arms share no reachable point under the as-built mount, identified a buildable mount correction, and that correction was applied and re-verified. Ten-repeat, known-answer verification ran throughout, catching twenty-six measurement defects before they entered this report as findings. And a first pilot ran under ethical approval with four operators across three tasks, showing shorter completion times and fewer re-grips under assistance with no measured cost in hand travel or responsiveness.

### 6.2 What was not achieved

No result here was validated on a physical Kinova arm carrying a load. The full two-person study – operator and wearer, measured together under the same autonomy – was not run; the pilot substitutes a smaller, operator-only comparison, made possible once VR controllers offered a route around the master’s damaged channels. The planning report’s haptic objective was not delivered, and is not deliverable through this platform’s control interface without a hardware change.

### 6.3 Further work

In priority order: repairing the two identified master channels, which would make the mannequin master usable as the direct-control baseline the full study needs; calibrating the wrist-camera-to-arm transform, the largest unmeasured term in the positioning budget and the reason no open-loop grasp has yet closed on hardware; correcting the clearance check to sample link geometry rather than link origins, the one item on this list that is a safety defect rather than an incompleteness; and running the full two-person protocol of [Chapter J](#) with a separate wearer, now that ethical approval and a working input path both exist, to measure what this project was ultimately built to ask

– the exchange rate between what shared autonomy buys an operator and what it costs the person carrying the arms.

## **6.4 Closing**

The most transferable outcome of this work is not the mapping, the safety architecture, or the pilot's result. It is the discipline that produced them: twenty-six times a result that looked like a system failure turned out to be a defect in the measuring tool, and the habit of establishing which one is in hand before acting on it is the part of this project most worth carrying into the study it was built to enable.

# Appendix A

## The master arm

The master is a miniature mannequin reproducing the Gen3's joint sequence at roughly a third of its scale, moved with the fingers of one or both hands. Source: `firmware/master_arm/teensy_final.ino`.

### A.1 Mechanical chain and electronics

Each arm carries seven potentiometer-instrumented joints, alternating a roll about the preceding link's own axis with a bend perpendicular to it; link lengths sum to 0.272 m. A roll joint moves the tip only through a downstream bend (rotating a straight arm about its own axis translates the tip by zero), which is why the loss of J2 or J4 costs reach while J3 or J5 costs almost nothing.

Master arm, from CAD — zero configuration

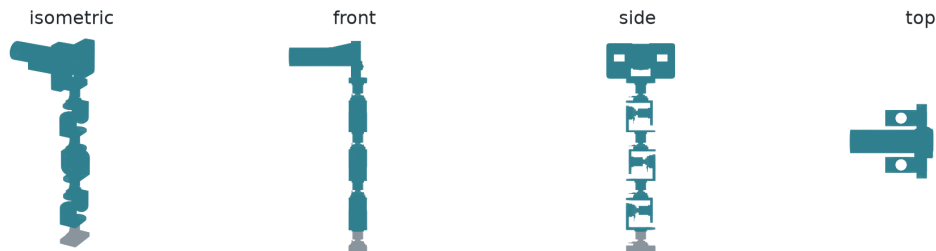


Figure A.1: Four orthographic views of the assembly at the zero configuration. The mechanical model carries seven potentiometer housings, a shaft coupling at each joint, and a wrist platform for the inertial sensor and force-sensitive resistor.

A Teensy 4.1 reads fourteen potentiometers, two MPU-6050 inertial sensors, two force-sensitive resistors and two buttons, emitting one ASCII text frame per cycle at 100 Hz ([Table A.1](#)). An earlier firmware revision placed two potentiometer channels on A4/A5, which are physically the default Wire bus both inertial sensors use; calling `analogRead()` on a pin the I<sup>2</sup>C peripheral is driving corrupts the bus intermittently and returns meaningless potentiometer values, with no symptom that looks like a pin conflict from the outside. The fix excludes both pins from analog use: Teensy 4.1 exposes eighteen analog-capable pins, minus two leaves sixteen, exactly what fourteen potentiometers plus two force sensors require – no spare.

Table A.1: Serial frame, in transmission order.

| Field        | Count | Format | Meaning                             |
|--------------|-------|--------|-------------------------------------|
| k1j1. .k1j7  | 7     | %.1f   | left arm joints, 0 to 360°          |
| k2j1. .k2j7  | 7     | %.1f   | right arm joints                    |
| K1IMU, K2IMU | 6+6   | %d     | $a_x a_y a_z g_x g_y g_z$ , raw LSB |
| fsr1, fsr2   | 2     | %d     | grip force, raw counts              |
| btn1, btn2   | 2     | %d     | latched toggle state                |

## A.2 Signal conditioning

Each potentiometer channel is read at 12-bit resolution and passed through an integer exponential moving average,

$$e_{n+1} = \frac{\alpha r_n + (256 - \alpha)e_n}{256}, \quad \alpha = 64, \quad (\text{A.1})$$

a smoothing factor of 0.25 per sample. The filtered count maps to degrees by a clamped linear transform that discards the outer 50 counts at each end of the ADC range, where a potentiometer wiper is least linear.

Recorded captures show each potentiometer producing a new *distinct* value at only 14.7 to 17 Hz against the 100 Hz frame rate, so 82 to 88 % of consecutive rows repeat – every per-row statistic computed before this was noticed was invalidated by it ([Chapter G](#)). At the power-on default of  $\pm 2$  g full scale, accelerometer sensitivity is 16 384 LSB/g; the widely copied value of 8192 LSB/g corresponds to the  $\pm 4$  g range and, used here, halves every reported acceleration – invisible in the elevation estimate (a uniform scale error cancels in a normalised vector) but fatal to the accelerometer gate, which then rejects every frame and looks like a dead sensor. Force-sensitive resistors are read as raw integers with no filtering; the mapping applies its own deadband and hysteresis downstream, so the discrepancy between the design (which specifies a filtered value) and the firmware in hand is recorded rather than resolved.

## A.3 The CAD is not the kinematic source

Neither STEP file contains a single kinematic-joint entity; they are solid geometry for manufacture, describing one fixed configuration. No URDF or xacro anywhere in the workspace references either file – **the master arm has no URDF at all**. What the software uses is seven measured constants in `srl_teleop/master_calibration.py`, taken directly from the physical arm. The CAD’s joint-module pitch agrees with those constants to 1.2 %, which is a cross-check, not a licence to treat the CAD as the source: it cannot distinguish the 43 mm roll from the 37 mm bend within one measured pitch.

# Appendix B

## The route the design took

*A record of process, not of result: the routes tried and abandoned before [Chapter C](#) took its delivered form. Recovered prior programs are archived at `thesis_report/_source/prior_work/`, dated 7 June to 27 July 2026; the full divergence analysis against the planning report is at `thesis_report/DIVERGENCE.md`.*

The first master-to-slave mapping sent master joint angles directly to slave joint angles. It failed for a structural reason rather than a tuning one: forward kinematics compounds every joint angle in the chain, so a 9 mm/3° disagreement between the master’s CAD and its measured link lengths ([Section A.3](#)) propagates to a materially wrong slave pose at the wrist, and there is no reconstruction path once a channel carrying an intermediate joint fails. Task-space mapping, sending master *tip displacement* to slave end-effector displacement, decouples the two chains and is what shipped – at the cost of leaving the slave’s redundant degree of freedom unspecified, which is what [Chapter I](#) later shows to be expensive near the wearer.

Two further redirections are worth recording because each was forced by a measurement rather than chosen for elegance. The planning report’s haptic objective – physical springs at the master, virtual impedance at the slave – was abandoned once the arm’s cyclic control interface was found unusable over the available network path ([Chapter 4](#)): an impedance law running at 18 Hz is not an impedance law. And validating the master-slave mapping in simulation before touching hardware, as planned, used MuJoCo [[TET12](#)] rather than Isaac Lab [[NVI26](#)] – it starts in under a second, its model format is one readable file, and for a kinematic-mapping question its contact solver is irrelevant, so the heavier alternative bought nothing.

Sensor faults turned out to be the dominant failure mode, not control faults: injecting a stuck analog reading sent a wrong angle to the corresponding slave joint with no reconstruction available under the joint-space mapping, which is the finding that motivated the graceful degradation architecture of [Chapter C](#) and, with it, the reduced azimuth/elevation/reach decomposition that the task-space mapping made possible.

# Appendix C

## System as built

Chapter 3 gives the command mapping and safety architecture in summary. This appendix covers what is not repeated there: the package architecture, the full mode registry, the desk arrangement the laboratory actually operates, calibration, the operating window, and the verification audit.

### C.1 Architecture

The workspace is six packages with a one-way dependency graph (Fig. C.1). Direct teleoperation is the baseline condition of every planned experiment, so if the teleoperation package imported anything from the autonomy package the comparison would be circular; verified both by source inspection and by the inverse-kinematics follower still reaching its running state with the autonomy and perception packages deleted from the install tree.

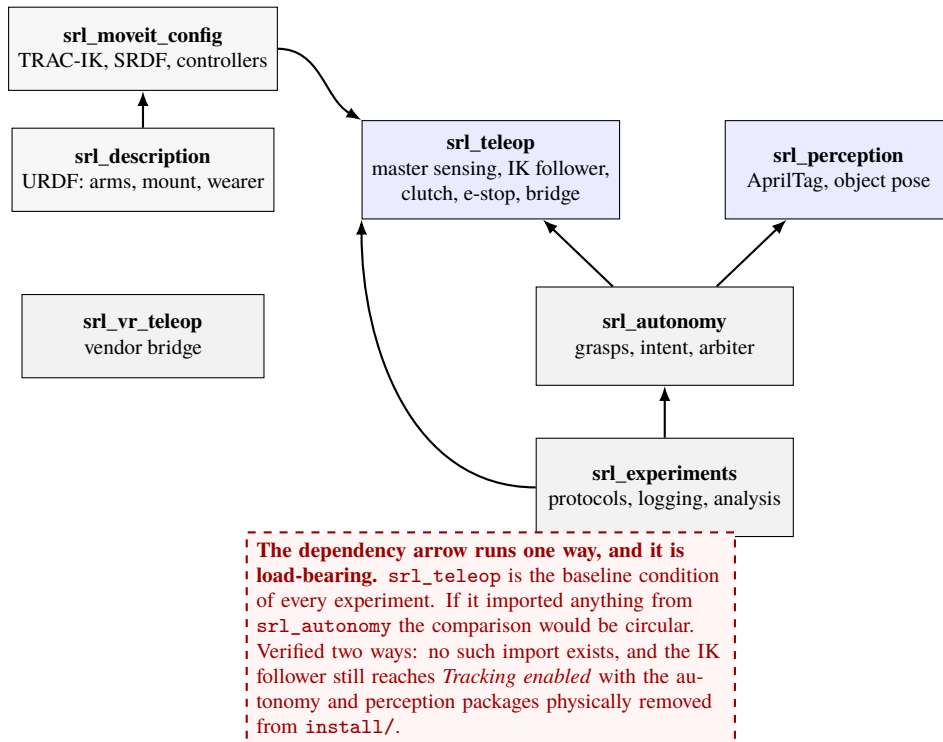


Figure C.1: Package decomposition and the one-way dependency graph.

## C.2 Command mapping, in full

The commanded position is built in spherical form about the master’s shoulder rather than by forward kinematics from seven joint angles, each component taken from whichever sensor observes it most directly:

$$\phi = \arcsin\left(\hat{a} \cdot \frac{\mathbf{g}}{\|\mathbf{g}\|}\right) \quad \text{elevation, from gravity} \quad (\text{C.1})$$

$$\psi = \theta_1 - \theta_1^0 \quad \text{azimuth, from the shoulder roll} \quad (\text{C.2})$$

$$r = \|\text{FK}(\theta_1, \theta_2, \theta_3, \theta_4, 0, 0, 0)\| \quad \text{radius, magnitude only} \quad (\text{C.3})$$

with  $\hat{a}$  the arm’s long axis in the sensor’s own frame, captured once at rest, and  $\mathbf{g}$  the measured acceleration. Elevation is mount-independent and drift-free; radius is dominated by the bend joints and tolerates a dead roll passed through as zero. The commanded end-effector position is

$$\mathbf{p} = \mathbf{p}_{\text{anchor}} + s(\mathbf{d} - \mathbf{d}_{\text{ref}}), \quad (\text{C.4})$$

with  $\mathbf{d}$  the master tip in the robot frame and  $\mathbf{d}_{\text{ref}}$  the reference latched at clutch engagement, from the first *valid* frame after boot at the master’s actual pose. An earlier version initialised the reference to the origin of the master’s own frame, so the first command was a step of the master’s entire tip vector (roughly 0.22 m) in whatever direction the master happened to be lying.

### C.2.1 Collision avoidance in three stages

The wearer is in the collision model: thirteen collision elements were added, paired with thirty-six exclusions for the mount-adjacent proximal links (the arm bases are bolted to the harness and permanently adjacent, so without exclusions every pose is trivially in collision – everything distal of the first arm link stays enabled, which is what actually protects the wearer). Avoidance is graduated: collision-aware inverse kinematics rejects a configuration; on rejection the redundant degree of freedom is re-seeded and the solve retried; a geometric clearance floor measured from the transform tree provides the final margin. The floor is measured against the *fused* body of [Section D.4](#), not the semantic collision description, which excludes forty-four proximal pairs and would report a pose valid with the upper-arm tube inside the person; and `base_link` sits 0.1610 m from the torso and is moved by no joint, so a clearance reading of exactly that value is a measurement of the mount, not the arm ([Chapter I](#)). Driving the commanded pose directly at the wearer’s head and torso stopped the arm at 0.072 to 0.079 m with zero clearance-floor blocks, so the earlier stages did the work and the floor remained a backstop.

## C.3 Control modes, in full

Orientation assist is a stub: it would supply wrist orientation under an orientation constraint, and the inverse-kinematics plugin in use ignores the constraint field entirely (270 solver calls, success identical with and without it). It needs a constraint-aware plugin or explicit yaw sampling.

### C.3.1 The desk arrangement

The laboratory’s actual arrangement seats the operator across the room, facing the wearer, holding the VR controllers as a six-degree-of-freedom input device and watching the physical robot directly. **Nobody wears the**

Table C.1: Every delivered operating mode. The numbering is 01, 02, 03, 04 and 06: there is no 05, and the gap is preserved because recorded data carries these labels.

| Mode                            | Input                | Robot decides                  | $v_{\max}$ (rad/s) |
|---------------------------------|----------------------|--------------------------------|--------------------|
| 01 direct, mannequin            | master arm           | nothing                        | 0.60               |
| 02 direct, controllers          | tracked controllers  | nothing                        | 0.60               |
| 03 shared autonomy              | master, vision       | wrist orientation              | 0.60               |
| 04 shared autonomy, controllers | controllers, vision  | wrist orientation              | 0.60               |
| 06 full autonomy                | typed or spoken goal | target, approach, grasp, place | 0.15               |

**headset**; it stands on a shelf as the tracking reference for the controllers. The alignment between the operator’s frame and the robot’s is a measured yaw and can never be a mirror: facing someone and copying their motion is a reflection (determinant  $-1$ ), which looks correct in position alone but reverses every commanded orientation. The angle is measured by having the operator move 0.4 m toward the robot, with a perpendicular cross-check; calibration refuses to write a value if the two motions are not perpendicular and horizontal.

Bringing the WebXR transport up for the first time (2026-08-19) exposed two defects no unit test would have found: the bridge node had never started at all, because an attribute named `clients` shadowed a read-only property of the base class; and its axis map had been written for the conventional robotics frame rather than this project’s, so a hand moved 3 m forward commanded 3 m to the wearer’s right. The delivered route serves the client page and the wss socket from one origin so a single certificate exception covers both, needs no adb or sideloading, and was verified end to end on 2026-08-19.

## C.4 Calibration

## C.5 The operating window

Every capability reaches one window: a feature reachable only from a terminal does not exist for the person running a session with a wearer standing in the rig. `scripts/verify_gui_buttons.py` presses every control and checks what the press did (257 checks). The first complete run found four defects: five session controls raised an exception inside a Qt slot, which the framework swallows, so they did nothing and said nothing; and the stray-process preflight for the one-stack constraint had never executed, because its module was never imported.

## C.6 Verification methodology

The rule – a surprising failure is evidence about the instrument until the instrument has been cleared – and the ten-repeats-over-a-densified-path practice are stated in [Chapter 3](#); [Chapter G](#) is the full case-by-case record.

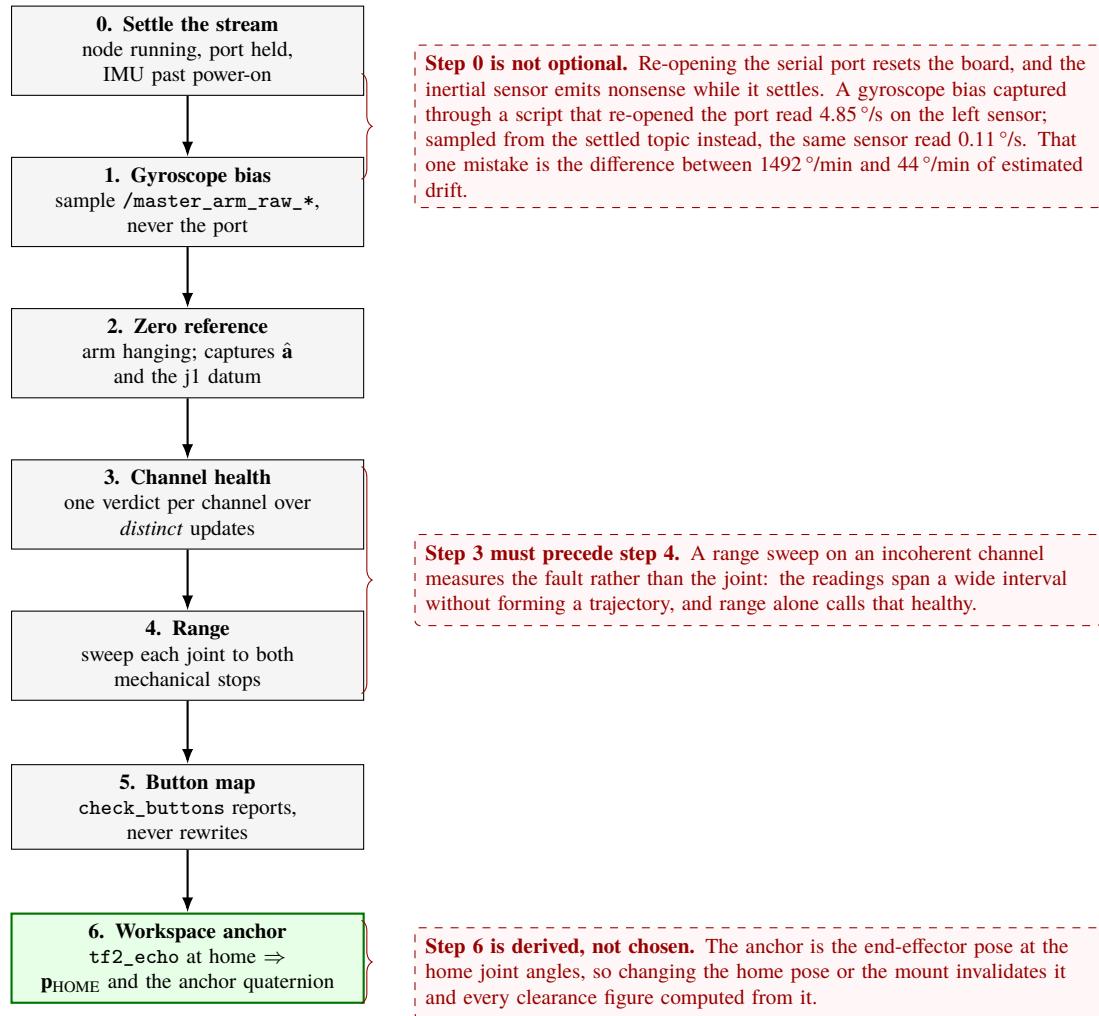


Figure C.2: The calibration procedure, in order. A calibration routine measures whatever the instrument was actually doing at the time, so establishing that the instrument is in a valid state is part of the measurement and not a preliminary to it.

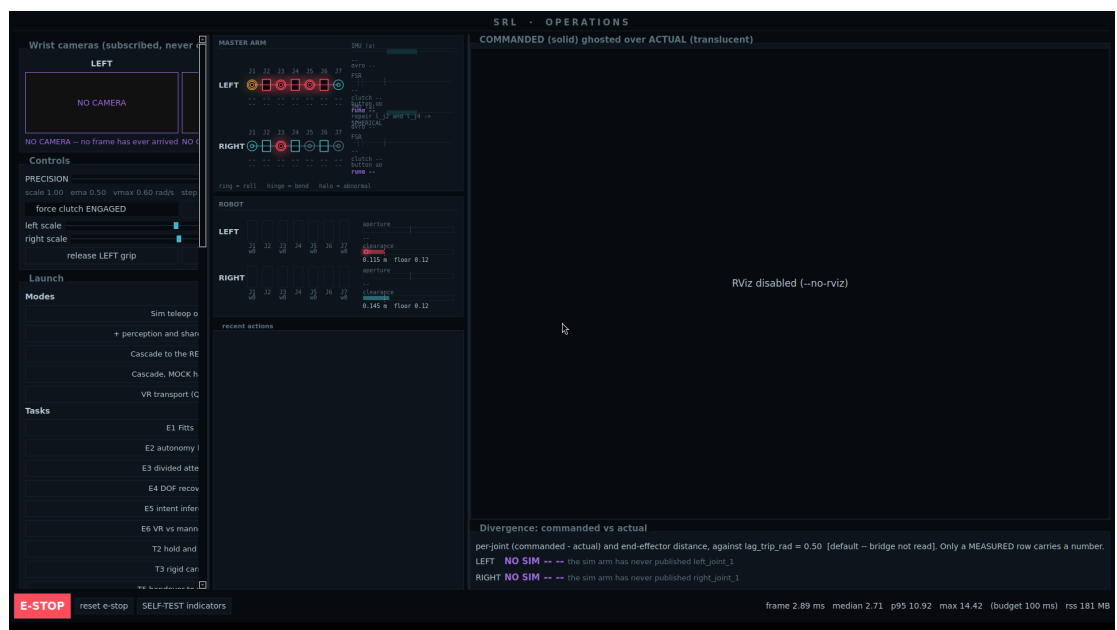


Figure C.3: One command opens it. Left: set-up, run, status. Centre: mode registry and per-mode controls. Right: the three-dimensional commanded view beside a panel drawn from the arms' own reported joint states, with the divergence between them read out beneath both, so a mismatch between commanded and reported state is visible without reading a log.

## Appendix D

# Perception, environment mapping and autonomous manipulation

Chapter 3 summarises this stage. Almost everything in the delivered task layer previously ran on *declared* coordinates verified at ten repeats over densified paths, none of which survives a different table, a different object, or the same object displaced by 50 mm. Three components capable of sensing the same quantities already existed and had never been composed: a support-plane finder used only to print a description, a voxel-world planner constructed nowhere outside its own self-test, and a work-surface module whose depth-driven height setter nothing called.

### D.1 Detection and grasp synthesis

Detection is open-vocabulary (YOLO-World [CSG<sup>+</sup>24] with FastSAM [ZDA<sup>+</sup>23]), falling back to a hue-and-depth filter; the backend that produced a detection is reported with it. Range is used as a discriminator where hue cannot be: the laboratory’s teal robots and the target green block differ in hue by only two units, well inside noise, but sit 2.8 m away against a 1.6 m working range, so range separates them at zero extra cost – the same size-at-range expression ( $w = sZ/f$ ) also rejects a 210 mm pad presented as a 40 mm cube. Grasp synthesis takes the principal axes of the object’s depth cloud: writing the mean-centred points as  $\mathbf{P}_k$ , the axes are the eigenvectors of the scatter matrix

$$\mathbf{C} = \frac{1}{N} \sum_{k=1}^N (\mathbf{P}_k - \bar{\mathbf{P}})(\mathbf{P}_k - \bar{\mathbf{P}})^\top, \quad \mathbf{C}\mathbf{u}_j = \lambda_j \mathbf{u}_j, \quad \lambda_1 \geq \lambda_2 \geq \lambda_3, \quad (\text{D.1})$$

and the jaw closes across  $\mathbf{u}_3$ , the direction of least extent, capped at

$$w = \max_k \mathbf{u}_3^\top \mathbf{P}_k - \min_k \mathbf{u}_3^\top \mathbf{P}_k \leq w_{\max} = 85 \text{ mm}. \quad (\text{D.2})$$

A geometric method can *name* the cause of a refusal (too wide, no depth, too few points), which a learned alternative cannot and which also would not fit beside the detector in 4 GiB of video memory.

A reachability gate checking candidate grasps for reach and wearer clearance was inoperative for the life of the module: it scored the clearance from the whole kinematic chain including the immobile mount stub, which reads a constant 0.2202 m in every pose and so could never fail against the 0.15 m floor. It now reads the moving-chain clearance (0.4031 m at home) and a regression test requires a named refusal when aimed at the wearer’s torso.

## D.2 Measuring the working area

The sweep holds one wrist attitude per pass and translates, printer-bed-probe style, rather than re-aiming at each grid point: re-aiming would make a residual camera-to-wrist error a *different* error in every frame, which can neither cancel across views nor be estimated from them. A frame is accepted only once no joint has moved by more than 0.0015 rad over 400 ms – a tenth of the arrival tolerance.

Table D.1: The fused map after one full sweep, against the scene it was measuring.

| Quantity                    | Measured                              |
|-----------------------------|---------------------------------------|
| Surface height              | 1.250 26 m against a declared 1.250 m |
| Objects recovered           | 6 of 6, every centre within 1 mm      |
| Object extents              | every dimension within 3 mm           |
| Coverage of reachable cells | 98.65 %, 0 stillness refusals         |
| Elapsed                     | 13 min 15 s                           |

An earlier surface error of  $-5.2$  mm was itself an instrument defect (Chapter G): the reported height was the fitted plane’s intercept at the *origin*, a point inside the wearer, not its height over the working area. Reachability (a fact about the robot, computed once) and the scene map (a fact about the room, rebuilt from scratch every time) are separate products with separate lifetimes; `scripts/pick_from_map.py` refuses to move from a map older than thirty minutes, because a stale map is indistinguishable from a fresh one by inspection.

## D.3 Planning and natural language

`joint_planner.py` implements RRT-Connect [KL00] in joint space (the binding constraint is the whole arm’s distance to the wearer, not the hand), evaluated against the same moving-chain clearance the guard enforces, with every edge densified at 0.05 rad to match the path checker. Without a planner, a pick whose pre-grasp and grasp poses were each individually clear passed within 0.1462 m of the wearer mid-path.

Table D.2: Seventy-five phrasings, delivered grammar against an earlier revision checked out of version control.

| Outcome                     | Earlier  | Delivered |
|-----------------------------|----------|-----------|
| Correct                     | 21       | <b>34</b> |
| Asked a clarifying question | 14       | 14        |
| Refused                     | 37       | 27        |
| <b>Misunderstood</b>        | <b>3</b> | <b>0</b>  |

The three earlier misunderstandings shared one mechanism: a compound instruction (*put the blue ones on the blue mat and the green ones on the green mat*) was cut at the first destination cue, so the two-target check counted targets only in the head clause, planned the first, and silently discarded the second while reporting success.

## D.4 Observing the wearer

The clearance floor of Chapter 3 is enforced against a body. `wearer_tracker_node.py` replaces the mannequin in the robot description, part by part, with one built from markerless pose estimation and metric landmarks lifted by perspective-n-point [LTN<sup>+</sup>19]: 28 to 34 ms inference, 61.6 ms median frame-to-estimate, 62 to 562 ms end to

end (the spread is a configured publication rate limit, not the estimator). The safety argument is one monotonicity rule enforced in one function: **the camera may only ever make the wearer larger** – per body part, `fuse()` keeps whichever of the tracked and mannequin primitive is *nearer* the robot. A body injected 0.10, 0.30, 1.00, 5.00 m further away moved the enforced margin by nothing; six injected faults (dark frame, blank frame, two people, empty frame, occluded arms, a control) were refused seven of seven, each naming the fault. The honest limitation is on the fallback side: if the wearer raises their arms and tracking is then lost, the guard reverts to arms-down and believes 0.317 m of clearance where there is 0.093 m.

Before this was connected, the tracker reached the planning scene and *not* `mount_guard_node`, which builds its wearer at import time – the body the planner avoided and the body the floor enforced against were two different bodies. A second camera observing the workspace from outside the arms is implemented, refuses to republish stale frames, and both calibration tools pass self-tests to 0.43 % focal length and 0.0000 m/0.000° extrinsic error – but **no camera has ever been attached to it**: the device is the host’s own webcam, and forwarding it into the Linux environment needs an administrator-level bind that was refused.

# Appendix E

## Motion generation and simulation-to-hardware transfer

[Chapter 4](#) reports the headline numbers. This appendix gives the mechanism.

### E.1 Why a per-joint clamp cannot stay on the line

The limiter it replaced applied, per cycle and independently to each joint  $i$ ,  $q_i^{(k+1)} = q_i^{(k)} + \text{clip}(q_i^* - q_i^{(k)}, -\Delta, +\Delta)$  with one step bound  $\Delta$  shared by all seven joints. Writing the required displacement as  $\mathbf{e} = \mathbf{q}^* - \mathbf{q}^{(k)}$ , staying on the straight joint-space line requires the applied step be *parallel* to  $\mathbf{e}$ ; the clamp applies  $\text{clip}(\mathbf{e}, -\Delta, +\Delta)$  component-wise, which is parallel to  $\mathbf{e}$  only when every component is inside the bound or every component saturates equally. In every other case the direction of travel rotates and the end effector leaves the path. This is why sweeping  $\Delta$  over a factor of 35 does not drive the excursion to zero – it settles at 68 mm – and why the excursion is a property of the algorithm, not its tuning.

### E.2 What replaced it

Ruckig [\[BK21\]](#): time-optimal, jerk-limited, per-degree-of-freedom limits, accepting a non-zero target velocity. Phase synchronisation drives every joint from one scalar phase  $\sigma(t) \in [0, 1]$ :

$$\mathbf{q}(t) = \mathbf{q}^{(k)} + \sigma(t)(\mathbf{q}^* - \mathbf{q}^{(k)}), \quad \sigma(0) = 0, \sigma(T) = 1, \quad (\text{E.1})$$

so the commanded configuration is on the segment between endpoints by construction for every  $t$  – unlike merely stretching each joint’s own profile to the slowest joint’s duration (time synchronisation), which measures 9.5 mm of path deviation on the same case against phase’s 0.02 mm. The acceleration and jerk limits are *assumed* and labelled so: `joint_limits.yaml` declares zero acceleration limits for all fourteen joints, so there is nothing to read; only the velocity limits are the robot’s own.

Five defects were caught in the replacement, four of which would have passed a conventional test suite: re-planning every cycle destroyed phase synchronisation (a mid-profile re-plan cannot reproduce the profile it is half through); floating-point noise forced a re-plan every cycle even after fixing that, because the target was recomputed via a wrapped subtraction not bit-identical to itself; the synchronised-clamp fallback reported zero acceleration (because it does not model one) and so passed a continuity check by fabricating the quantity being checked; the off-path metric had a 0.1 mm grid-resolution floor and was reporting the instrument’s resolution as

the robot’s behaviour; and an arrival test used an absolute tolerance on travels spanning a factor of twenty-five.

### E.3 The simulation-to-hardware gap

Of 252 recorded per-joint errors, 211 lie between  $0.25^\circ$  and  $0.30^\circ$  with the sign following the direction of travel – a *bound*, not a spread, which identifies a terminal offset rather than noise. Two candidate models, a terminal offset and a proportional gain error, coincide at one travel distance and differ by a factor of five at 20 mm; propagating the recorded errors through the manipulator Jacobian reproduces the measured Cartesian error to 90 % ( $r = 0.923$ ) under the offset model and not the gain model. The data had been recorded for weeks in `recordings/baselines/arm_directional_calibration.json` before anything read it.

An earlier report claimed the terminal error is speed-independent across a fivefold range; the velocity limit was read once at bridge start-up and seven of thirty-six runs moved *faster* than commanded, so the three speeds were three replicates of one condition and the speed dependence has never actually been measured. The bridge deadband was  $1.0^\circ$  against a measured  $0.305^\circ$  park – a factor of 3.3 – and is  $0.10^\circ$  now, with an opt-in overshoot correction; neither has been tried on hardware.

### E.4 The positioning error budget

$$\sigma_{\text{RSS}} = \sqrt{\sum_i b_i^2}, \quad \sigma_{\text{worst}} = \sum_i b_i, \quad (\text{E.2})$$

with  $\sigma_{\text{worst}}$  the honest figure against the 30 mm capture gate, since the two largest terms are declared constants rather than noise. Corrections – the pad offset re-derived from forward kinematics (13.45 to 0.05 mm), the shell bias removed via the measured support plane, and the terminal-offset overshoot (joint term 7.24 to 1.88 mm) – take the budget from 18.64 to 12.91 mm RSS and 33.39 to 19.99 mm worst case, *provided* the camera-to-wrist transform is small. [Section H.7](#) measured it directly and it is not.

### E.5 The reaction budget

Table E.1: Limb travel during the measured wearer-tracking latency against the 150 mm clearance floor. Speeds are literature ranges, labelled as assumptions; latencies are measured.

| Limb motion      | Latency | Travel        | Against the floor |
|------------------|---------|---------------|-------------------|
| Deliberate reach | 62 ms   | 50 to 93 mm   | 62 % of it        |
| Deliberate reach | 562 ms  | 450 to 843 mm | exceeds it        |
| Quick reach      | 62 ms   | 93 to 155 mm  | exceeds it        |
| Startle          | 62 ms   | 155 to 248 mm | exceeds it        |

In five of six modelled limb motions the limb travels further than the entire margin before the guard is aware anything has changed – [Chapter 5](#) treats this as the principal open safety question.

## Appendix F

### The vision pipeline, measured on the real cameras

*Measured on four physical cameras, 2026-08-30 – together with [Section H.7](#), the only hardware numbers in this report.*

The pipeline is instrumented at thirty-three stages per camera; every stage draws itself on the frame it was given, states its formula, and reports its quantities with units, so a claim can be checked rather than taken on trust. A stage has three possible outcomes, not two: it succeeds, it *refuses* and names its cause and remedy, or it is *absent by design* (the webcam’s fifteen depth stages, because it has no depth sensor) – rendering an absence identically to a defect has cost this project findings before.

#### F.1 The orientation defect

Three places in the software independently decided whether the scene depth camera is mounted upside down, and disagreed. Settled by correlating each frame’s grey signature against the HD webcam (rotated by nothing, on the same side of the room) in both orientations: the un-rotated frame agrees at  $r = 0.66$  to  $0.70$  over four captures, the rotated one at  $r = 0.27$  to  $0.34$  – the same verdict every time, a factor of 2.4 apart. **The camera is mounted the right way up**, and every earlier scene figure was inverted. Because the wrong rotation had been applied *consistently* to the image and the intrinsics together, the recovered point cloud was an exact reflection of the true one and every distance, extent and residual was preserved – the defect was invisible to every metric the pipeline itself computes and was found only by comparing one camera’s picture against another’s.

#### F.2 What the cameras measured

Table F.1: One frame per camera. The inlier fraction decides whether the row below it means anything.

|                          | left wrist | right wrist | scene depth |
|--------------------------|------------|-------------|-------------|
| pixels returning a depth | 89.4 %     | 80.5 %      | 14.5 %      |
| plane inliers            | 74.0 %     | 52.8 %      | 3.8 %       |
| plane residual, RMS      | 1.43 mm    | 1.59 mm     | 3.41 mm     |
| objects posed in 3D      | 5          | 12          | 12          |
| graspable within the jaw | 0          | 0           | 0           |

**The wrist cameras measure a work surface to about 1.5 mm** – within a factor of three of the same code against a renderer, the first evidence the surface-fitting stage transfers to hardware. **The scene camera does not identify a support plane**: its best plane holds 3.8 % of the points, so every quantity that depends on it (height,

object boxes) rests on a plane one point in twenty-six agrees with. **No object in any frame could be grasped:** the nearest object's narrowest axis measures 144 mm against an 85 mm jaw. **The open-vocabulary detector returned nothing on any camera** for the prompt *green cube*; the classical colour-and-geometry path found the same objects it did not.

Two defects in the first runs were software, not hardware. The pick path's own depth working band, correct for a wrist camera, applied unchanged to a camera across the room retained **0 of 85 061 returns** – the band is now declared per source. And two stages disagreed about the same frame because depth at  $480 \times 270$  forward-projected into colour at  $1280 \times 720$  lands one sample in roughly every seven pixels, so the above-plane mask was a dot pattern that failed a connectivity check while a per-mask stage saw it fine; splatting each projected sample raised coverage to 64.4 % and 58.8 %.

The arms were to be parked at the pick pose for this session; measured rather than assumed, the left arm was  $10.67^\circ$  and the right  $13.53^\circ$  from the reference configuration, against a  $2^\circ$  tolerance, so no figure from this session is captioned "at the pick pose."

# Appendix G

## Measurement validity: the full record

*Chapter 5 states the validity argument and the two numbers that carry it. This appendix is the evidence: what each of the twenty-six cases actually was, grouped by mechanism.*

**A surprising failure is evidence about the instrument until the instrument has been cleared.**

Twenty-six times a measurement in this project was wrong and the system was not. In every case the wrong reading was *plausible* – none announced itself – and two would have removed results now central to this thesis. Sorted by mechanism, they fall into five groups.

**An absence read as a value (7 cases).** Zero variance from oversampling a slower sensor ([Section H.2](#)); a detection rate of 0 to 4 % from a renderer producing images unlike anything the detector was trained on; a surface height reporting the fitted plane’s intercept at the origin rather than over the working area.

**A test harness constructing the condition it tests (6 cases).** A reachability control comparing a search-cap lower bound against a boundary, which was never going to pass regardless of the arm’s true reach; a path-verification failure from the verifier’s own test path running in the one direction that leaves the feasible region.

**State left over from a previous run (5 cases).** A reachability sweep measured with the arms parked 0.87 and 0.59 rad from home, reporting zero failures under the same name as the genuine measurement.

**Order of operations inside the measurement (5 cases).** An isotropy statistic taken as the minimum over directions of a single sweep, dominated by rare early solver terminations and varying by a factor of two between runs; stable once repeated ten times and read as a median.

**A constant asserted in more than one place, measured in none (3 cases).** The scene camera’s mounting orientation ([Section F.1](#)); the wrist-to-pad offset, treated as a scalar when it is a function of gripper opening.

Two numbers carry the argument. **Fifteen of the twenty-six were caught by an independently known quantity contradicting the reading** – five by inspecting the code, six by looking at the raw artefact, usually an image. Care was not the defence that worked; redundancy was. **Fourteen of the twenty-six were already answerable from something in hand** – a published stroke, an arithmetic prediction, a quantity exact by construction – which is the recommendation this project’s experience supports: build the cross-check in rather than trying harder.

## G.1 What follows from it

Every reachability figure in this report carries its repeat count, because one solver call is one coin flip. Every stage of the simulation campaign carries controls that must *fail*. Where a quantity has never been measured, the text says so rather than offering an estimate – the camera mounting transform in the positioning budget is one such term, and it is not small ([Section H.7](#)). Any unreproduced single measurement in this report should be read as provisional.

## G.2 Three standing caveats

**Simulation is not the arm.** Every quantitative result except the channel measurements and the pilot comes from simulation with mock hardware, whose force channels are identically zero – any quantity derived from force is structurally undefined, not merely unmeasured. **Latency figures are lower bounds**, measured between software stages, excluding the operator, the display and the physical arm’s response. **The one-stack constraint:** two instances of the master node once split the serial stream between them and invalidated a full day of measurements before the cause was found; this is now prevented at the file-descriptor level, but any measurement taken before that prevention carries a risk that cannot be retrospectively excluded.

# Appendix H

## Results, in full

[Chapter 4](#) reports the headline numbers with their evidence class. This appendix is the supporting record: every table the simulation campaign, the hardware sessions and the safety and instruction-handling measurements produced, with only enough text to say what each is.

### H.1 The simulation campaign

`scripts/run_simulation_campaign.py` runs every simulation stage by one command and records what each produced with its exit status and duration; `scripts/make_sim_figures.py` refuses to emit a table whose stage did not run. Sixteen stages, all of which ran; 13 927 inverse-kinematics calls at zero failures.

Table H.1: The simulation campaign inventory.

| Stage                       | Needs              | Ran | Time     |
|-----------------------------|--------------------|-----|----------|
| <i>computer vision</i>      |                    |     |          |
| cv_arithmetic               | standalone         | ✓   | 12 s     |
| cv_orientation              | standalone         | ✓   | < 1 s    |
| cv_replay                   | standalone         | ✓   | 88 s     |
| <i>teleoperation</i>        |                    |     |          |
| teleop_budget               | standalone         | ✓   | < 1 s    |
| teleop_motion               | standalone         | ✓   | 5 s      |
| teleop_orientation_cost     | simulated platform | ✓   | 20.6 min |
| teleop_replay               | recorded data      | ✓   | < 1 s    |
| teleop_smoothing            | standalone         | ✓   | < 1 s    |
| <i>robot tasks</i>          |                    |     |          |
| tasks_accuracy              | standalone         | ✓   | < 1 s    |
| tasks_dance                 | simulated platform | ✓   | 38 s     |
| tasks_msc                   | simulated platform | ✓   | 48 s     |
| tasks_status                | standalone         | ✓   | < 1 s    |
| tasks_t1                    | simulated platform | ✓   | 2.2 min  |
| <i>autonomy</i>             |                    |     |          |
| autonomy_arbitration        | standalone         | ✓   | 4 s      |
| autonomy_instructions       | standalone         | ✓   | < 1 s    |
| <i>study design</i>         |                    |     |          |
| study_power                 | standalone         | ✓   | 17 s     |
| <i>transfer to hardware</i> |                    |     |          |
| sim_to_real                 | recorded data      | ✓   | < 1 s    |

### H.1.1 Computer vision against a constructed scene

Twenty-one known-answer checks on a frame whose plane, object pose and size were chosen. The fitted plane normal lands  $0.0001^\circ$  from the constructed one; a 40 mm cube measures  $40.8 \times 40.7 \times 40.7$  mm and centres to 0.41 mm. Two naive shortcuts are measured wrong by name: an axis-aligned bounding box reads 41 % too wide, and taking the object centre as the mean of its visible points is out by 16.15 mm along the surface normal (a single view sees a shell, not a solid) – corrected to 0.35 mm by assuming the object rests on the fitted plane.

### H.1.2 Shared autonomy: what the intent estimator is sensitive to

Sweeping the declared-position error over the range a real table spans, a declared-coordinate estimator becomes *confidently wrong* 50 % of the time once the declaration is one object-pitch out; a measured-coordinate one stays at 0.0 % across the same sweep, with the correct rate flat at 71.7 to 75.7 %. The correct rate never exceeds 76 % in any condition, including perfect information – the remaining quarter is ambiguity, and the arbiter is specified to stay in direct teleoperation whenever ambiguous.

### H.1.3 Full autonomy: seventy-five phrasings

Correct interpretations rose from 21 to 34 of 75 against an earlier grammar revision, refusals fell from 37 to 27, misunderstandings fell from 3 to 0. Ten of fourteen clarifying questions carry a scored reply; all ten resolve correctly.

## H.2 Channel health

*Measured on recorded hardware data, 2026-08-06.* Six of fourteen channels met the alive criterion.

Table H.2: Per-channel verdicts, over distinct sensor updates.

| Left arm |                     | Right arm |                    |
|----------|---------------------|-----------|--------------------|
| Channel  | Verdict             | Channel   | Verdict            |
| j1       | intermittent, 2.2 % | j1        | alive, $195^\circ$ |
| j2       | incoherent, 5 %     | j2        | alive, $261^\circ$ |
| j3       | incoherent, 8 %     | j3        | incoherent, 14 %   |
| j4       | incoherent, 6 %     | j4        | alive, $291^\circ$ |
| j5       | incoherent, 12 %    | j5        | dead               |
| j6       | incoherent, 8 %     | j6        | alive, $250^\circ$ |
| j7       | alive, $26^\circ$   | j7        | dead               |

Left j7 had been recorded as railed and dead: the evidence was a spread of exactly zero over forty consecutive *transmitted rows*, which is what an oversampled sensor (82 to 88 % repeated rows) produces regardless of health. Regressing true reach on the surviving channels plus the inertial sensor gives  $R^2 = 0.133$  (left) and  $R^2 = 0.986$  (right), naming the repair as exactly left j2 and j4.

### H.3 Axis correctness

Regressing commanded pose against the master’s physical spherical position (6443 samples,  $R^2 = 0.999$ ) found a gain matrix of  $\text{diag}(+1, -1, -1)$  against the intended identity: raising the master drove the arm down. Corrected. The lateral axis remains wrong and cannot be fixed by any axis map: measured directions meant to be  $90^\circ$  apart are  $137.7^\circ$  apart, and a rotation preserves angles, so no rotation maps one triad onto the other. Gyroscope-derived azimuth halves the residual on the left arm ( $25.6^\circ$  to  $13.5^\circ$ ) but is not enabled: the right sensor drifts at  $44^\circ/\text{min}$  stationary.

### H.4 Safety: fault injection and timing

Fourteen of fourteen injected faults handled: channel dropout, board disconnect/reconnect, emergency stop mid-motion, wrong-object latching, ambiguous-target flicker, stale object, detection failure, logging failure, identifying data in a manifest, non-monotonic clock, session loss, zero detections, network dropout. Dead-man timing against a 1.00 s deadline: healthy master, 243 frames, no trip; **degraded (publishing at full rate, frozen source timestamp): trip at 0.94 to 0.99 s** – the case that could not have passed before keying the dead-man on the source timestamp; clean unplug: trip at 0.97 s. A separate emergency-stop defect: the trip handler called a blocking real-driver shutdown first, so decision at 0.97 s produced trip *logged* at 4.98 s – fixed by parking first, from local publishes only.

### H.5 Task-set verification

Every coordinate of the five-task set verified at ten repeats over paths densified to 20 mm, both arms confirmed at 0.0000 rad from home before any sample: zero failures, and the result reproduces field for field on re-run.

Table H.3: Five-task verification.

| Task            | Coverage                            | Result       |
|-----------------|-------------------------------------|--------------|
| Positioning     | 3 targets per arm                   | 3/3, 3/3     |
| Pick and place  | 3 picks per arm, approach and place | 3/3, 3/3     |
| Rigid carry     | 3 paths, 25 densified waypoints     | all verified |
| Compliant carry | as rigid carry                      | all verified |
| Dual pursuit    | 26-direction shell per arm          | 26/26, 26/26 |

### H.6 Performance figures

*Control-path latency; a lower bound on end-to-end latency.*

The Kortex figure constrains the architecture: the cyclic control interface needs a 1 kHz loop on a dedicated link, and over this machine’s virtualised network each cyclic write costs about 10 ms – at 100 Hz the write alone consumes the cycle budget and no command reaches the arm while telemetry keeps arriving. The high-level velocity interface avoids this because it holds a commanded motion between sends.

Table H.4: Measured rates and latencies, against mock hardware.

| Quantity                | Value                       | Note                      |
|-------------------------|-----------------------------|---------------------------|
| Master frame rate       | 100 Hz                      | firmware loop             |
| Distinct sensor updates | 14.7 to 17 Hz               | <a href="#">Chapter A</a> |
| VR packet rate          | 68.5 Hz                     | 822 packets / 12.0 s      |
| VR round trip           | 8.66 ms median, 16.5 ms p95 |                           |
| Voice stop to /estop    | 2 ms                        |                           |
| Kortex high-level loop  | 18.4 to 18.7 Hz             | two round trips/cycle     |

## H.7 The first hardware session

*Measured on a physical arm – the only section of this chapter whose numbers come from a moving Kinova arm.* A single arm was driven from its own wrist camera for the first time. Three findings survive as results. **The camera-to-wrist transform is wrong by roughly  $8.5^\circ$  and rotates with the wrist:** the surface normal measured from four poses of a stationary table read  $9.74, 15.42, 16.26, 18.14^\circ$  from vertical, a worst pairwise disagreement of  $8.45^\circ$  – 34 mm at 0.23 m, 59 mm at 0.40 m. **The depth sensor itself is excellent:** the support plane fits at 94.8 % inliers, 0.46 mm residual, and a 40 mm calibration cube measures  $39.2 \times 38.9 \times 40.9$  mm – only the transform out of the camera frame is wrong. **The middleware was a first-class hazard:** shared memory in a state where no new process could join while running nodes continued normally, and an unset middleware variable caused silent discovery failure. Both present as an unresponsive robot.

## H.8 Verification, in totals

Table H.5: What is checked automatically, as of 2026-08-31.

| Measure                                      | Count                  |
|----------------------------------------------|------------------------|
| Automated tests passing                      | 1505                   |
| Test modules                                 | 136                    |
| Checks on the operating window               | 257                    |
| Faults injected and handled                  | 14 of 14               |
| Simulation stages, all of which ran          | 16                     |
| Inverse-kinematics calls, failures           | 13 927, 0              |
| Known-answer checks on the vision arithmetic | 21                     |
| Simulated studies of the analysis procedure  | 20 000                 |
| Recorded demonstration cells                 | 17, at 8 camera angles |
| Vision stages instrumented on real cameras   | 132                    |

## H.9 What is not measured

Anything derived from a force reading (the mock returns identically zero); object detection rate at working distance (the renderer used to measure it is out of the detector’s training distribution); grasp success against a physical object; end-to-end latency including the arm and operator; how the terminal joint error varies with commanded speed (retracted in [Chapter E](#): three speeds were three replicates of one condition); speech recognition against a human speaker (no audio input device on this host; every voice figure characterises the grammar, not a recog-

niser); two arms driven simultaneously on hardware (measured at 8.1 to 12.2 Hz with send latency spiking to 519 ms, past the bridge's own watchdog).

# Appendix I

## Workspace characterisation and its consequences

Chapter 4 gives the headline finding and the mount fix it produced. This appendix is the chain of measurements behind it.

### I.1 The two arms share no reachable workspace

Over a 63-cell grid spanning the frontal working volume, arms verified at home, no cell was reachable by both arms. The result survives freeing the wrist yaw (raising each arm's individual reach from 16/18 to 21/21 cells, still 0/63 shared) and additional solver retries.

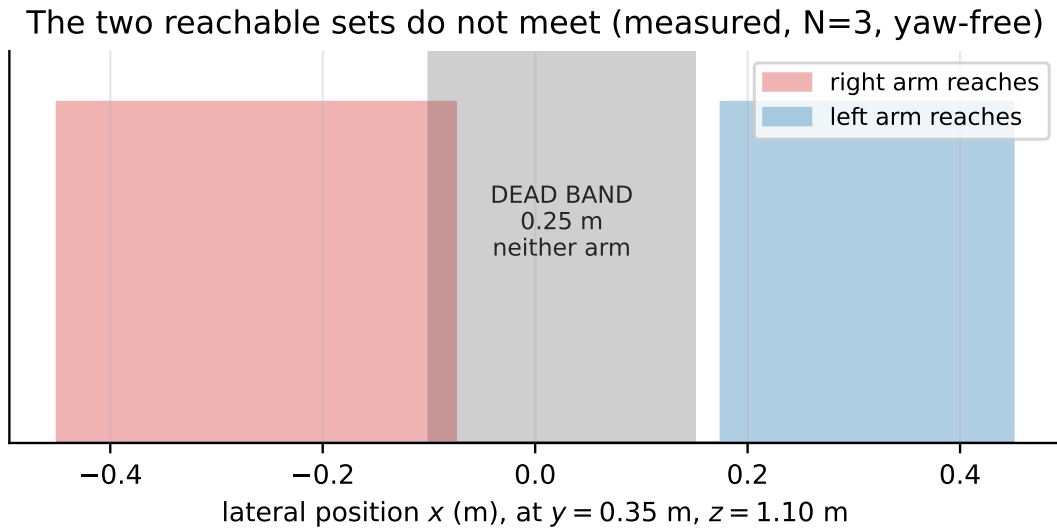


Figure I.1: The two reachable sets do not meet. Measured at  $y = 0.35$  m,  $z = 1.10$  m, yaw free, three repeats per point: the right arm reaches  $x \leq -0.10$  m, the left  $x \geq 0.15$  m, a 0.25 m dead band between.

This holds within  $x \in [-0.6, 0.6]$ ,  $y \in [0, 0.4]$ ,  $z \in [0.90, 1.30]$  m, with the as-built mount and the legacy home joint angles – the arms share no workspace *as mounted and as parked*, and the cause is not the mount's rotation (proved invariant) but the arms' asymmetric home joint angles, which are ground truth because the simulation-to-hardware bridge replays them onto the real arm.

### I.1.1 The fix is buildable

A parametric sweep moving the mounts inboard, forward and down, with additional forward tilt, raises shared cells from 3 to **22 of 63** within a buildable 0.20 m mount separation (splaying the mounts outward *hurts*). Recorded and deliberately not applied at the time (changing the mount invalidates the home pose, the anchor and every clearance figure), then applied later as one deliberate pass with the whole task set re-measured:

Table I.1

| Quantity                             | Before                        | After               |
|--------------------------------------|-------------------------------|---------------------|
| Clearance at home                    | 0.1610 m                      | 0.2202 m            |
| Nearest robot part to wearer at home | the mount (no joint moves it) | the moving chain    |
| Task objects resting on the surface  | not achievable                | achieved            |
| Pick-and-place targets off centre    | 0.530/0.350 m                 | 0.290 m either side |

## I.2 One boundary, two independent constraints

Wearer clearance and top-down grasp feasibility, measured for different reasons, land on the same lateral boundary: clearance rises monotonically and crosses the 150 mm floor at 0.25 m; top-down grasp candidates fail entirely (0/9) at  $|x| = 0.25$  m and succeed entirely (9/9) at  $|x| = 0.30$  m, with collision checking disabled too – kinematic infeasibility of the wrist orientation, not obstruction.

## I.3 What the pinned wrist costs at the work point

Reach under each orientation policy, walked from home, understates the cost badly because home is in free space. Walked instead from (0.4, 0.175, 1.12 m) on the working surface, with the 150 mm floor enforced throughout:

Table I.2: Mean reach under each orientation policy. The two columns disagree by a factor of seven, because reach is meaningless without the point it was measured from.

| Orientation policy               | from home (m) | from work point (m) |
|----------------------------------|---------------|---------------------|
| pinned to the anchor (delivered) | 0.460         | <b>0.065</b>        |
| within a 15° cone                | 0.492         | 0.304               |
| within a 45° cone                | 0.530         | 0.325               |
| position only                    | 0.547         | 0.363               |

At the point where the work happens, the delivered policy leaves the arm 65 mm of mean reach against 363 mm with the wrist free – a factor of 5.6. The mechanism is redundancy, not the wearer: no walk from the work point stops on the clearance floor, they stop because there is no solution – a fixed six-degree-of-freedom pose on a seven-degree-of-freedom arm spends the entire redundancy, and pinning the wrist removes the joint that would otherwise move the elbow clear while the hand stays put. Freeing the roll alone buys 0 mm at the work point; almost all recovery comes from the first 15° (80 % of the pinned-to-free gap), which the follower's own 15° solving cone already supplies – the reason the task set solves at all.

Table I.3: Four attempts to place usable work within 100 mm of the body centre line, at ten repeats.

| Question                                          | Answer                                                                 |
|---------------------------------------------------|------------------------------------------------------------------------|
| Innermost workable column                         | $ x  = 0.325 \text{ m (L), } 0.450 \text{ m (R)}$                      |
| Same, wearer's arms deleted entirely              | 0.300/0.375 m; every column 0.125 to 0.300 m binds on the <b>torso</b> |
| Objects resting on a surface, 3360 configurations | <b>no configuration</b> within 100 mm of centre                        |
| Same, top-down approach                           | <b>0 of 840</b> cells, any height 0.70-1.10 m                          |

## I.4 The centre of the working space, asked four ways

The constraint is the torso, not the wearer's limbs: sweeping five arm postures moves the boundary by at most 75 mm, and two postures (arms clasped behind the back, arms out to the sides) place the wearer's own limbs inside the clearance floor at home, before the robot does anything. Neither arm crosses the centre line at all – zero of ten solutions, both arms, every cross-side slot – which is why the pick-and-place task draws the *side* and lets colour follow it.

## I.5 Consequences for the task set

Inter-arm handover is impossible: no control, planning or autonomy change can produce a point neither arm reaches. A bimanual task must span the dead band or exploit simultaneity. The coupled carrier's original 310 mm width placed each grip inside the dead band (10 of 25 waypoints failed once the arms were verified at home); widened to 500 mm it satisfies both boundaries, and the carrier's sag tolerance was re-derived to keep the failure threshold inside the reachable band rather than outside it, where it would measure nothing.

# Appendix J

## Protocol for the two-person study

*This is the study [Chapter 3](#) says the system was built for. The pilot in [Chapter L](#) is a smaller, operator-only precursor; this protocol – with a separate wearer – is still the target.*

### J.1 The question and hypotheses

Does supplying part of the operator’s control with autonomy improve performance, and what does it cost the person carrying the arms? No prior study measures both under the same autonomy in the same trial ([Chapter 2](#)). Five hypotheses, stated with direction before any data exists: task completion time is shorter under shared autonomy (H1); operator workload is lower (H2); **wearer-reported trust is lower under shared autonomy** (H3), against the direction of H1/H2, taken from a concealed-operator study [[ZGD+26](#)] and tested here with autonomy that is genuinely a program and a wearer with a concurrent task; the wearer’s sense of agency is lower while ownership is unchanged (H4), since the two dissociate [[LSK+08](#)]; cross-arm interference is lower under shared autonomy while phase lag is not (H5), which would identify the benefit as attentional. **H3 and H4 are the hypotheses worth running the study for**; H1 and H2 exist to establish the platform can show an effect at all.

### J.2 Design

Two modes (direct teleoperation, shared autonomy) crossed with five tasks, within subjects, sixteen dyads, roles fixed for the whole session. Two modes rather than four is costed, not a convenience: four modes across five tasks needs 260 to 320 min against a wearer exposure cap of 48 min for carrying 17 kg continuously; a between-subjects design across modes needs ninety-six participants for the same sensitivity, since between-subject variance here exceeds the effect sought. Mode order is counterbalanced with Williams squares, not a Latin square, because a Latin square balances position but not immediate sequence, and a task with a learning curve is sensitive to the sequence.

**Grasping under direct teleoperation is not achievable:** the default orientation mode pins the wrist to the anchor, the pick task needs a 169.7° rotation from it, and no master channel can command that. T2 (pick and place) therefore carries no mode contrast and is analysed as a single condition – retained because grasp success under autonomy is worth measuring, and because it is the one task where autonomy’s advantage over teleoperation on this platform is categorical rather than quantitative.

Table J.1

| Task               | Measures                                                                   | Mode contrast        |
|--------------------|----------------------------------------------------------------------------|----------------------|
| T1 positioning     | Throughput vs. index of difficulty [Fit54]; single- vs. both-arm decrement | both modes           |
| T2 pick and place  | Grasp success, placement accuracy                                          | shared autonomy only |
| T3 rigid carry     | Coordination error as object tilt                                          | both modes           |
| T4 compliant carry | Coordination error as gripper separation                                   | both modes           |
| T5 dual pursuit    | Cross-arm interference under divided attention                             | both modes           |

### J.3 Measures and procedure

NASA-TLX [HS88] after every block, both participants separately (the two roles load differently, and the comparison between them is the object of the study); Borg CR10 [Bor82] exertion from the wearer between every block, since a single end-of-session rating turns a covariate into a guess; trust and embodiment [LSK<sup>+</sup>08] at the end of session, wearer only for embodiment. A safety observer, present throughout, holds an emergency stop and watches the wearer rather than the screen; their presence is enforced in software as well as procedurally – the safety node treats two seconds of heartbeat silence as withdrawal of consent to move. Consent is taken separately, in separate rooms.

### J.4 Ethics and risk

The dominant risk is a 17 kg manipulator pair, moving under partly autonomous control, within reach of a seated person’s head and torso. Mitigations: the 150 mm margin enforced over the whole path and measured geometrically (Section C.2.1); a stop within the wearer’s own reach and a second held by the observer; twenty minutes of carried-load exposure at a time with exertion rated between blocks; order counterbalanced against fatigue confounding condition; consent restated at every break; identifying data refused in code (`write_manifest()` raises on name, email, date of birth, address or telephone number). Video recording is not part of the protocol: a video of a participant wearing this rig is not anonymous, and no measure here requires one.

## Appendix K

### The analysis procedure, and its behaviour on constructed data

The analysis [Chapter J](#) would be evaluated with, measured against data whose answer is chosen. No dyad data exists yet; this is a property of the procedure, not a result.

The design is within dyads: for task  $j$ , the paired difference  $d_{ij} = y_{ij}^{\text{SA}} - y_{ij}^{\text{DT}}$  and its standardised mean  $d_z = \bar{d}_{\cdot j} / s_{d, j}$  is the quantity the design was powered for (distinct from the standardised difference of means, which deflates every effect by  $\sqrt{2}$  or more once within-dyad correlation is accounted for). Five tasks give five comparisons, corrected with Holm’s step-down procedure, which controls the family-wise error rate under any dependence structure and is uniformly more powerful than Bonferroni.

Applied to 20 000 simulated studies at the planned sixteen dyads (a true  $d_z = 0.73$  effect on all five contrasts; and separately, no effect at all), through the unmodified analysis code:

Table K.1:  $n = 16, m = 5, \alpha = 0.05$ .

| Constructed truth        | one contrast,<br>uncorrected | one contrast,<br>Holm-corrected | some contrast<br>in the family |
|--------------------------|------------------------------|---------------------------------|--------------------------------|
| $d_z = 0.73$ on all five | 0.777                        | 0.638                           | 0.973                          |
| no effect at all         | 0.052                        | 0.010                           | 0.050                          |

The null row is the check that could fail: a false-positive rate of 0.050 against nominal  $\alpha = 0.05$  shows the correction is implemented as specified. The design’s own 80 % power figure is confirmed *uncorrected* (77.7 %) but no contrast will be reported uncorrected; after Holm, recovering the effect on a *named* task falls to 63.8 %, and restoring 80 % on a named task needs twenty dyads, not sixteen. The study is nevertheless well powered for the question it actually asks – whether shared autonomy changes performance on this platform at all, where the family-wise power is 97.3 % – so a per-task null at  $n = 16$  must be reported as an interval, not an absence. Trial validity is defined in one shared module that exits non-zero when no trials survive or more than half are excluded; before that existed, a session with every trial invalidated analysed cleanly to empty means and undefined intervals, indistinguishable from a null result on inspection.

# Appendix L

## The pilot session, in full

[Section 4.4](#) reports the headline pilot numbers. This appendix is the full record: coverage, every session, and the supplementary measures.

Sixteen sessions were recorded on 2026-09-03 and 2026-09-04 (recordings/sessions/). Duration, hand-travel distance and lag come from the full-resolution position trace (20 Hz); clutch and emergency-stop counts come from each session’s event log. Participant identity is P1–P4 throughout, including in every figure pixel, consistent with `write_manifest()` refusing to record a name ([Section 3.5](#)).

Table L.1: Which participant performed which task under which condition. Coverage is uneven because a pilot session is limited by time.

| Participant | Object tracking           | Target reaching           | Position matching |
|-------------|---------------------------|---------------------------|-------------------|
| P1          | both (direct $\times 2$ ) | —                         | —                 |
| P2          | both                      | both (shared $\times 2$ ) | —                 |
| P3          | direct only               | shared only               | both              |
| P4          | both                      | —                         | both              |

Table L.2: Every recorded session. Duration includes setup and hesitation. Active share is the fraction of the session spent with the grip engaged. P1’s two direct-control sessions are a short warm-up followed by the recorded attempt, both retained as separate points.

| P  | Task              | Cond.  | Duration (s) | Travel (m) | Re-grips | Active |
|----|-------------------|--------|--------------|------------|----------|--------|
| P1 | Object tracking   | Direct | 147.4        | 0.31       | 19       | 50 %   |
| P1 | Object tracking   | Direct | 392.8        | 1.85       | 53       | 53 %   |
| P1 | Object tracking   | Shared | 114.9        | 1.50       | 12       | 61 %   |
| P2 | Object tracking   | Direct | 38.7         | 0.90       | 2        | 61 %   |
| P2 | Object tracking   | Shared | 169.6        | 1.57       | 25       | 76 %   |
| P2 | Target reaching   | Direct | 227.9        | 2.77       | 25       | 52 %   |
| P2 | Target reaching   | Shared | 78.6         | 0.70       | 11       | 65 %   |
| P2 | Target reaching   | Shared | 357.9        | 5.39       | 41       | 73 %   |
| P3 | Object tracking   | Direct | 617.4        | 7.09       | 52       | 83 %   |
| P3 | Target reaching   | Shared | 458.1        | 3.18       | 51       | 74 %   |
| P3 | Position matching | Direct | 315.9        | 3.72       | 35       | 67 %   |
| P3 | Position matching | Shared | 196.8        | 4.08       | 14       | 85 %   |
| P4 | Object tracking   | Direct | 195.6        | 2.07       | 37       | 68 %   |
| P4 | Object tracking   | Shared | 186.9        | 1.98       | 25       | 42 %   |
| P4 | Position matching | Direct | 171.9        | 1.65       | 12       | 30 %   |
| P4 | Position matching | Shared | 111.7        | 1.98       | 12       | 40 %   |

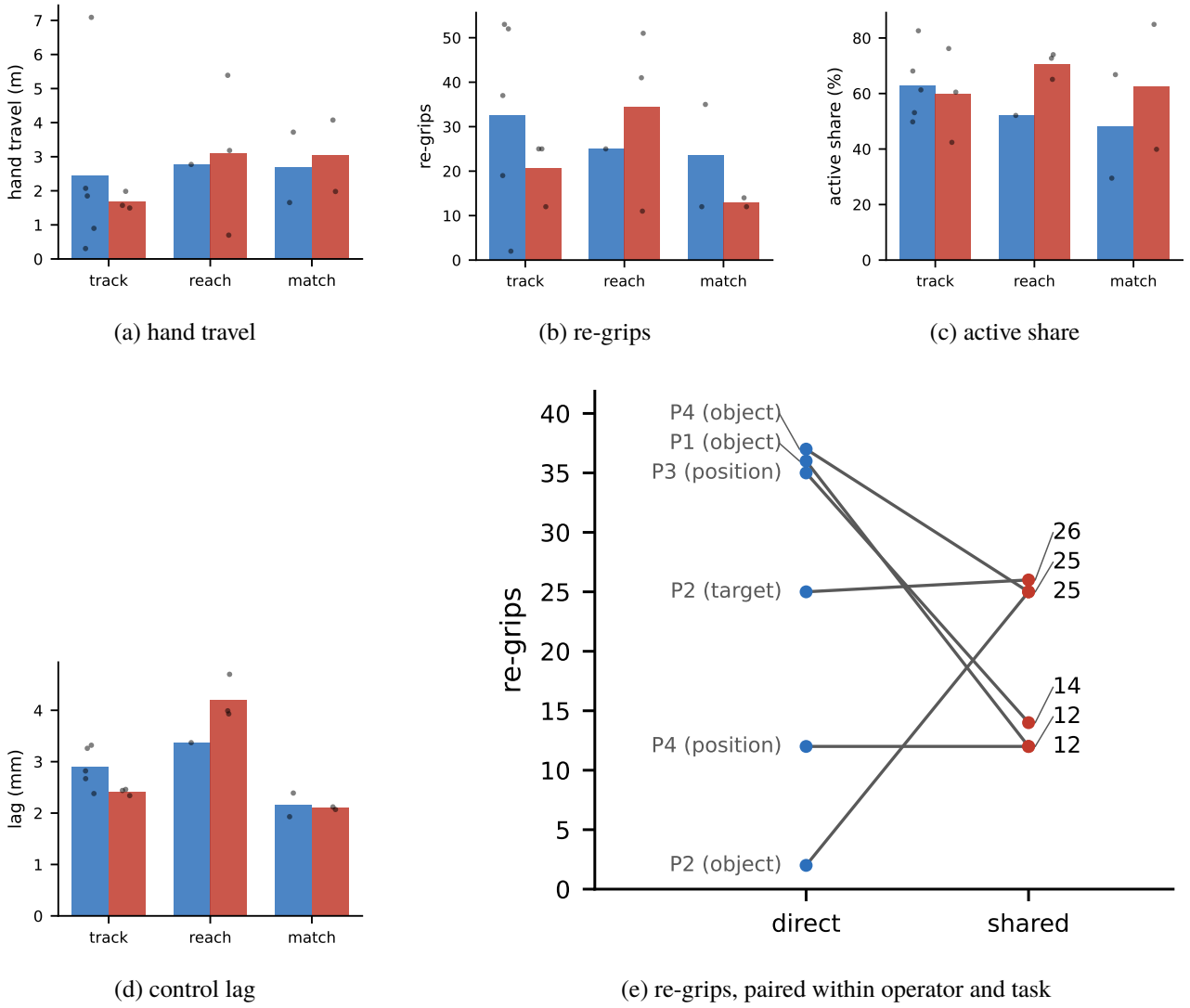


Figure L.1: The supplementary pilot measures, by task and condition. Distance and lag are essentially unchanged between conditions, which is why [Section 5.5](#) reads them as the more reliable measures in a pilot where session duration is confounded with setup time.

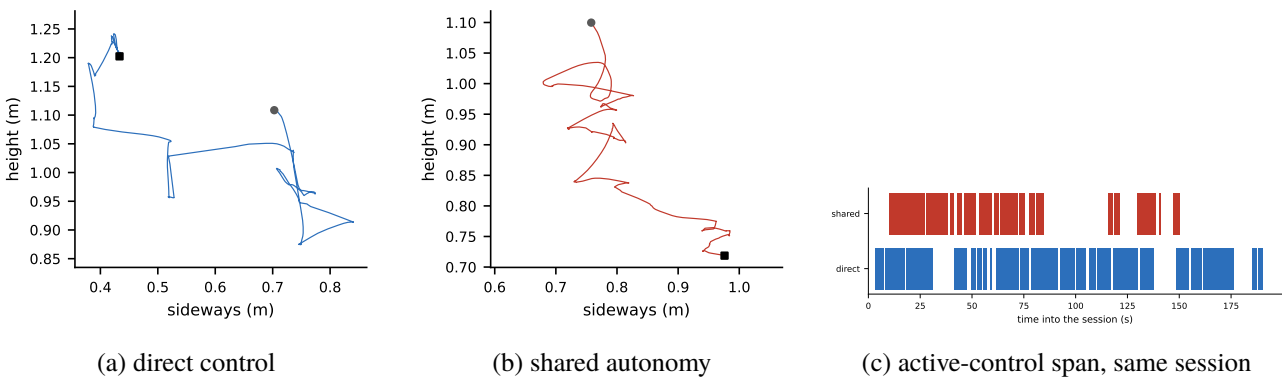


Figure L.2: The actual recorded end-effector path (`ee_left_x/y/z` in `trail.csv`) for one participant on the same task under both conditions, recorded back to back, with the corresponding span of active control alongside. Neither path is smooth; a shorter, straighter line would indicate a more direct approach to the target. Gaps in the engagement panel are the operator releasing the grip to reposition their hand before re-engaging.

## L.1 Sim-to-real tracking during the pilot

Every pilot session polled the physical Kinova arms’ own joint encoders alongside the simulated joint state, which makes a direct check possible: does the real arm actually follow the simulated one it was commanded through. [Figure L.3](#) is the best-tracking session across all sixteen (P4, object tracking, direct control, chosen by lowest sim-to-real joint RMS): the real arm’s joint 4 angle follows the simulated one to  $1.06^\circ$  RMS over 3859 samples, with the largest departures during the fastest motion and near-zero error while the arm holds still. This is the same terminal-offset mechanism [Section E.3](#) measures from replayed commands, seen here live during an operator session rather than in an offline replay – **it is evidence the sim-to-real bridge was live and tracking throughout the pilot**, not a claim that any pilot result was validated against a loaded arm, which requires the arm to be carrying and moving an object rather than only mirroring a joint angle.

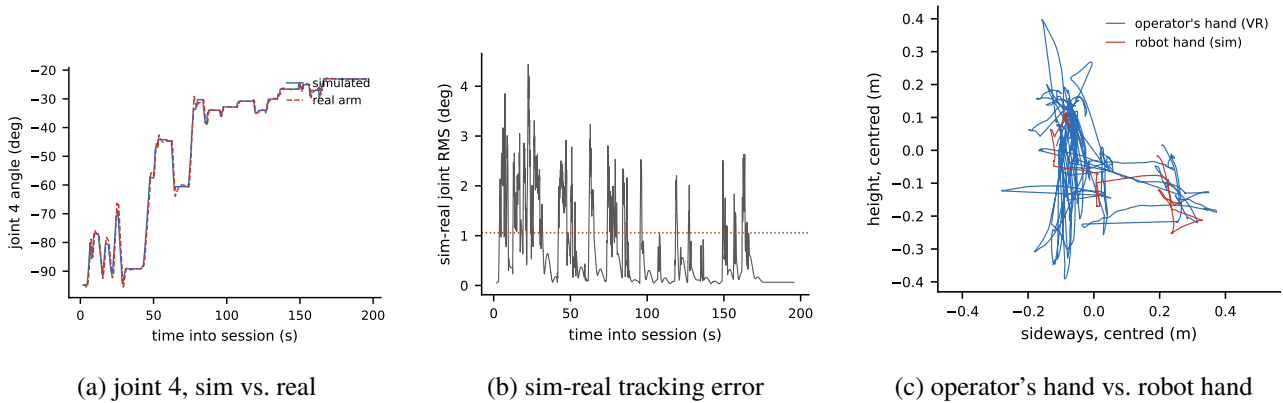


Figure L.3: The best-tracking pilot session by sim-to-real joint RMS. (a) One joint, simulated command against the real arm’s own encoder, over the session. (b) The resulting tracking error, root-mean-square over all seven joints; the dotted line is the session RMS,  $1.06^\circ$ . (c) The operator’s raw VR-hand path against the simulated robot hand’s achieved path, each centred on its own mean – the two live in different coordinate frames and at different scales by design, so an absolute overlay would not be meaningful; centring each on itself is the honest way to compare their shapes.

## L.2 The master-arm cohort

Three further operators (M1–M3, disjoint from P1–P4) attempted the same protocol from the instrumented mannequin master rather than VR controllers, across sixteen further recording attempts on 2026-09-02 and 2026-09-04. **Seven of those sixteen are confirmed empty:** `ros2 bag info` reports zero messages, or no bag metadata at all, for every attempt at the multimeter task and both of M1’s target-reaching attempts – the same recording-pipeline failure this project’s own session index for that day logged independently (eight of twenty-six recordings clean, ten with no data at all). The nine that did record are real sessions, 72 to 304 s each, tens of thousands of messages.

M1’s tracking error is not noise: it is consistent across all three of that operator’s sessions and separated from M2 and M3 by a factor of five to fifty. The mechanism this project’s own channel-health measurement predicts ([Chapter H](#)) is a session that leans on the incoherent left-arm channels rather than the coherent right-arm ones – all three of M1’s sessions used the *right* arm as the active hand ([Table L.3](#)), so a simple left/right split does not explain it on its own, and which specific channels M1’s sessions exercised has not been isolated. It is reported here as a finding this pilot surfaced, not a diagnosis: the master-arm modality can apparently produce sim-to-real

Table L.3: The nine recorded master-arm sessions. Joint RMS is the sim-to-real tracking error, computed the same way as Fig. L.3, over the active arm’s seven joints; see Fig. L.4.

| M  | Task, condition         | Duration (s) | Hand travel (m) | Joint RMS (deg) |
|----|-------------------------|--------------|-----------------|-----------------|
| M1 | Object tracking, direct | 154.1        | 21.44           | 21.2            |
| M1 | Object tracking, direct | 145.8        | 14.11           | 15.9            |
| M1 | (unspecified), shared   | 209.8        | 22.02           | 16.3            |
| M2 | Object tracking, direct | 110.9        | 10.85           | 2.2             |
| M2 | Object tracking, shared | 71.7         | 7.57            | 1.9             |
| M2 | Target reaching, direct | 303.8        | 20.80           | 1.8             |
| M2 | Target reaching, shared | 83.6         | 4.19            | 0.4             |
| M2 | Target reaching, shared | 206.4        | 11.60           | 3.0             |
| M3 | Target reaching, direct | 254.1        | 11.48           | 1.8             |

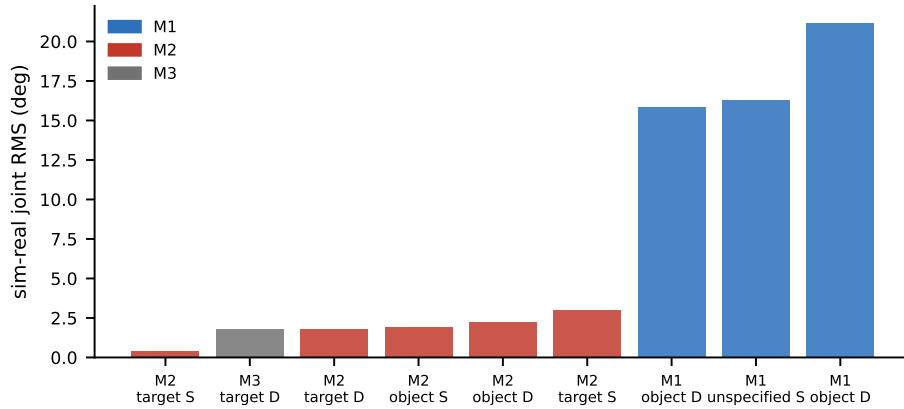


Figure L.4: Sim-to-real joint tracking error for every recorded master-arm session, sorted low to high. M2 and M3 track within  $3^\circ$  throughout; **every one of M1’s three sessions tracks an order of magnitude worse** (15.9 to  $21.2^\circ$ ).

tracking an order of magnitude worse than either VR teleoperation or another operator on the same hardware, on the same day, and the pooled VR-only comparison of [Section 4.4](#) would have missed it entirely.

**Limits.** Four participants with uneven task coverage is a pilot-scale sample: no significance test is reported, and none of the differences above should be read as a proven effect. The pooled figures mix three tasks together; the paired panels are the part of this pilot closest to the paired design of [Chapter J](#). Session duration runs from the start of the recording to the end, including any explanation or hesitation, so it is a session-length measure rather than a pure motor-execution time. Every session's emergency stop was engaged once, near the end, to stop the recording – a procedural marker rather than a safety incident – and is excluded throughout. This pilot has no separate wearer: every measure here is what the operator experienced, and the wearer-side hypotheses of [Chapter J](#) (H3, H4) remain untested.

# Appendix M

## Background, in full

*Chapter 2 states the gap this project addresses. This is the full review behind it, organised by theme, with each citation's scope stated precisely rather than implied.*

### M.1 Supernumerary robotic limbs

The field's founding demonstrations are Parietti and Asada's bracing limbs for aircraft fuselage assembly, later extended to whole-body support [PA16], which anchor against the environment so the reaction load leaves the wearer's body rather than passing through it (Fig. M.1). In the arrangement studied here the arms are free in space, so the chain closes through the wearer instead – and every reaction load is carried by a person who is not the person commanding the motion. This difference, not the control law, is what makes wearer clearance the binding constraint of Chapter I.

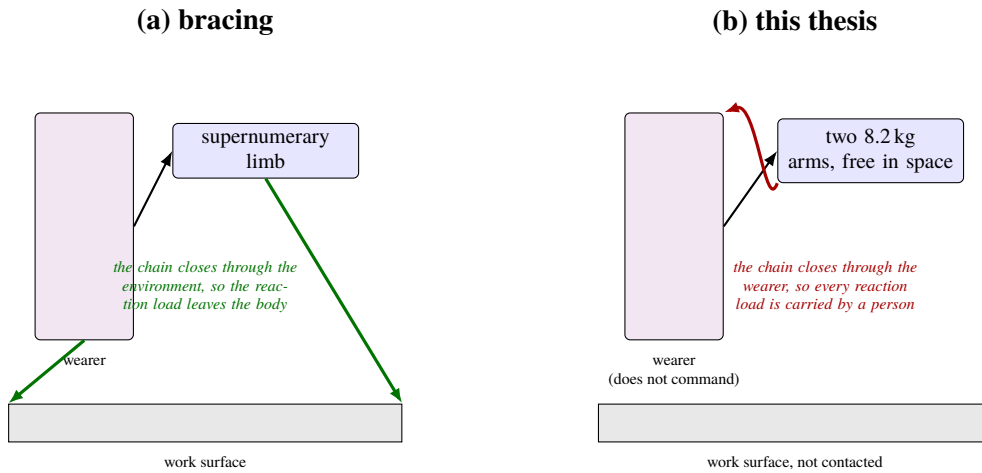


Figure M.1: Bracing (the field's founding configuration [PA14]) against free-space mounting (this platform).

Reviews since map the design space along consistent axes – mounting location trades reach against the moment arm imposed on the wearer, actuation trades bandwidth against worn mass [YHC<sup>+</sup>21, P<sup>+</sup>21, TL21, Z<sup>+</sup>23] – and a recent quantitative survey adds that the field reports embodiment far more often than task performance, and rarely both [Z<sup>+</sup>26]. Worn mass is the constraint reported least: each Gen3 masses 8.2 kg, the pair with harness exceeds 17 kg, and simulated wearer loads suggest the biomechanical cost is not negligible [MK24].

## M.2 The operator and the wearer as different people

Fusion mounts a robotic head and two arms on a worn frame, driven by a remote operator [SSM<sup>+</sup>18]. Two qualifications travel with that citation: its three modes (Direct, Induced, Enforced) are levels of *physical coupling* between the robot arms and the wearer’s own, not levels of machine autonomy – in all three the robot decides nothing – and the publication is a two-page demonstration with no user study and no performance numbers. It establishes that the architecture existed; it cannot support a claim about how well it worked.

Work on spatial guidelines for near-body interaction ran a concealed operator, varied apparent autonomy, and measured the wearer’s physiological arousal and self-reported trust [ZGD<sup>+</sup>26]: higher apparent autonomy made both worse. Their autonomy was a hidden human and their wearer had no task of their own, so the result does not settle what happens when autonomy is real and the wearer is occupied – it establishes only that the wearer’s experience is not a free variable. Disturbance rejection for the two-person case is partly solved: Zhang et al. compensate human-induced base motion and report end-effector errors of  $(1.37 \pm 0.58)$  mm [Z<sup>+</sup>24], with the human shoulder as a floating base – covering the case where wearer and operator are the same person (the base motion is partly predictable to the controller because the person commanding it also caused it), not the case studied here.

## M.3 Master interfaces for dissimilar kinematics

Guggenheim and Asada argue for miniature physical masters on the grounds that geometric similarity supplies feedback inherently, without a separate haptic display [GA21]. The argument covers similarity of *form*; this system’s master-to-slave reach ratio is roughly one to three, which is where it departs. Joint-space correspondence preserves posture but needs every master joint measured accurately; task-space correspondence decouples the chains at the cost of the slave’s redundant degree of freedom (Chapter B). Pinardi et al. review supplementary feedback in movement augmentation and find that feedback improving control does not reliably improve the sense of ownership [PLF<sup>+</sup>23] – a caution against treating either as a proxy for the other.

## M.4 Shared autonomy

Dragan and Srinivasa formalise arbitration as policy blending: a predicted user goal and the user’s actual input combined with a weight set by prediction confidence [DS13], explicit that blending aggressively on a wrong prediction is worse than not blending. Javdani et al. recast the problem as a POMDP over the user’s goal, approximated by hindsight optimisation [JSB15]: their user study found assistance that does not wait for high confidence lets operators finish faster with less input. Neither line of work was developed for a robot mounted on a second person – there, the cost of assisting toward the wrong goal is a manipulator moving toward someone who did not command it, not a slower trial.

## M.5 Measuring embodiment, workload and trust

The rubber hand illusion showed ownership can be induced over an object that is plainly not one’s hand [BC98]; Longo et al. decompose the resulting experience into ownership, agency and location, which can move independently [LSK<sup>+</sup>08] – an operator may feel authorship of a motion without feeling the arm is part of them, and a wearer may feel neither while bearing the consequences. Workload is conventionally the NASA Task Load

Index [[HS88](#)]; for a two-person system it must be administered to both, since the two will not load the same way. Physical exertion is better tracked with a Borg category-ratio scale [[Bor82](#)]. ANSI/ISA-101.01's central principle for operator displays [[Int15](#)] – colour is a scarce alarm channel, so a normal state should be largely uncoloured – transfers directly to a robot console.

## **M.6 The gap, restated precisely**

Fusion established the architecture and measured nothing. Shared autonomy raises operator performance where the robot is attached to nobody. Raising apparent autonomy lowers a wearer's trust where the autonomy is a concealed human and the wearer has no task. What has not been done is to measure both ends in one system, under the same autonomy, in the same trial – the engineering contribution of this thesis is a system in which that question could be asked; [Chapter 5](#) sets out what still stands between the system as built and the full session that would ask it.

# Appendix N

## Discussion, extended

*Chapter 5 is the argument. This appendix adds the detail that did not fit there.*

### N.1 The clearance model itself is not yet trustworthy

The most uncomfortable single finding of the hardware programme belongs here rather than in the results. Asked about a pose in which the arm was *physically against the mannequin*, one clearance model – the one that samples link *origins* rather than link geometry, and which runs on the physical arm during homing – returned 367.0 mm, the same value it returns for a visibly clear pose. Every clearance figure in this report computed through the guard’s capsule model stands (the majority); but the standing claim that wearer margin is enforced *at all times* was, at the time of writing, **not enforced in code on hardware**. This is the one item in [Chapter 6](#)’s further-work list that is a safety defect rather than an incompleteness, and it is placed above the mount change and above the study for that reason.

### N.2 Three things the real cameras changed

Beyond the headline numbers of [Chapter F](#): the surface-fitting stage transferring to hardware within a factor of three of its simulated figure is the one component of the perception chain now known to work outside simulation, and the one the pick path leans on hardest. The scene camera is a *different* instrument from the wrist cameras, not merely a distant one – an eighth of the depth returns, a twentieth of the plane support – and no autonomous behaviour should expect a support surface from it. And the learned detector contributed nothing the classical colour-and-geometry path did not; no result in this report rests on a learned detector, worth noting given that a language-model-free, learned-detector-optional design was argued for on cost grounds rather than accuracy grounds.

### N.3 Comparison against the planning report, in full

Divergences fall into three groups. *Superseded by measurement*: the haptic objective ([Chapter 5](#)). *Deferred and still open*: coordinated pick-and-place on the real platform – the task set is specified and verified geometrically and the runner exists, but the arms were not available for the period in which it would have run. *Added*: the safety architecture, the degradation architecture, the workspace characterisation and the pilot were not in the plan; the first two follow directly from discovering that sensor faults dominate, the third from asking whether the task set was reachable before running it.

# Bibliography

- [BC98] Matthew Botvinick and Jonathan Cohen. Rubber hands ‘feel’ touch that eyes see. *Nature*, 391(6669):756, 1998.
- [BK21] Lars Berscheid and Torsten Kröger. Jerk-limited real-time trajectory generation with arbitrary target states. *Robotics: Science and Systems XVII*, 2021.
- [Bor82] Gunnar A. Borg. Psychophysical bases of perceived exertion. *Medicine and Science in Sports and Exercise*, 14(5):377–381, 1982.
- [CSG<sup>+</sup>24] Tianheng Cheng, Lin Song, Yixiao Ge, Wenyu Liu, Xinggang Wang, and Ying Shan. YOLO-World: Real-time open-vocabulary object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [DS13] Anca D. Dragan and Siddhartha S. Srinivasa. A policy-blending formalism for shared control. *The International Journal of Robotics Research*, 32(7):790–805, 2013.
- [dWKB24] S. de Wolde, J. Kober, and R. Babuska. Current-based impedance control for interacting with mobile manipulators. *arXiv preprint arXiv:2403.13079*, 2024.
- [Fit54] Paul M. Fitts. The information capacity of the human motor system in controlling the amplitude of movement. *Journal of Experimental Psychology*, 47(6):381–391, 1954.
- [GA21] Jacob W. Guggenheim and H. Harry Asada. Inherent haptic feedback from supernumerary robotic limbs. *IEEE Transactions on Haptics*, 14(1):123–131, 2021.
- [HS88] Sandra G. Hart and Lowell E. Staveland. Development of NASA-TLX (task load index): Results of empirical and theoretical research. In Peter A. Hancock and Najmedin Meshkati, editors, *Human Mental Workload*, volume 52 of *Advances in Psychology*, pages 139–183. North-Holland, 1988.
- [Imp] Imperial College London. MUlti-limb Virtual Environment (Dr. Octopus). <https://www.imperial.ac.uk/a-z-research/human-robotics/dr-octopus-/>. Accessed April 2026.
- [Int15] International Society of Automation. *ANSI/ISA-101.01-2015: Human Machine Interfaces for Process Automation Systems*, 2015.
- [JSB15] Shervin Javdani, Siddhartha S. Srinivasa, and J. Andrew Bagnell. Shared autonomy via hindsight optimization. In *Robotics: Science and Systems XI*, 2015.
- [Kin22] Kinova Robotics. *Gen3 Ultra Lightweight Robot User Guide*, 2022.

- [KL00] James J. Kuffner and Steven M. LaValle. RRT-Connect: An efficient approach to single-query path planning. *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 995–1001, 2000.
- [LKK19] C. Lee, S. Kwak, and J. Kwak. Development, analysis, and control of series elastic actuator for a wearable robotic leg. *Frontiers in Neurorobotics*, 13:17, 2019.
- [LSK<sup>+</sup>08] Matthew R. Longo, Friederike Schüür, Marjolein P. M. Kammers, Manos Tsakiris, and Patrick Haggard. What is embodiment? a psychometric approach. *Cognition*, 107(3):978–998, 2008.
- [LTN<sup>+</sup>19] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, et al. MediaPipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*, 2019.
- [MK24] C. Moon and J. Kim. Assessing the physical impact of supernumerary limbs on a human subject: A simulation study. In *Proc. 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 1–4. IEEE, 2024.
- [NVI26] NVIDIA. Isaac lab documentation: Impedance control and motion generators. <https://isaac-sim.github.io/IsaacLab>, 2026. Accessed April 2026.
- [P<sup>+</sup>21] Domenico Prattichizzo et al. Human augmentation by wearable supernumerary robotic limbs: review and perspectives. *Progress in Biomedical Engineering*, 2021.
- [PA13] Federico Parietti and H. Harry Asada. Dynamic analysis and state estimation for wearable robotic limbs subject to human-induced disturbances. In *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, pages 3880–3887. IEEE, 2013.
- [PA14] Federico Parietti and H. Harry Asada. Supernumerary robotic limbs for aircraft fuselage assembly: Body stabilization and guidance by bracing. In *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, pages 1176–1183. IEEE, 2014.
- [PA16] Federico Parietti and H. Harry Asada. Supernumerary robotic limbs for human body support. *IEEE Transactions on Robotics*, 32(2):301–311, 2016.
- [PLF<sup>+</sup>23] Mattia Pinardi, Matthew R. Longo, Domenico Formica, Matija Strbac, Carsten Mehring, Etienne Burdet, and Giovanni Di Pino. Impact of supplementary sensory feedback on the control and embodiment in human movement augmentation. *Communications Engineering*, 2:64, 2023.
- [SSM<sup>+</sup>18] MHD Yamen Saraiji, Tomoya Sasaki, Reo Matsumura, Kouta Minamizawa, and Masahiko Inami. Fusion: full body surrogacy for collaborative communication. In *ACM SIGGRAPH 2018 Emerging Technologies*, pages 1–2. ACM, 2018.
- [TET12] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033. IEEE, 2012.
- [TGT<sup>+</sup>22] H. Toubar, G. Goldsmith, S. A. Tafrishi, M. Innala, and V. Bergstromer. Design, modeling, and control of a series elastic actuator with discretely adjustable stiffness. *Mechanism and Machine Theory*, 170:104720, 2022.

- [TL21] Yuchen Tong and Jianbin Liu. Review of research and development of supernumerary robotic limbs. *IEEE/CAA Journal of Automatica Sinica*, 2021.
- [YHC<sup>+</sup>21] Bo Yang, Jian Huang, Xu Chen, Caihua Xiong, and Yasuhisa Hasegawa. Supernumerary robotic limbs: A review and future outlook. *IEEE Transactions on Medical Robotics and Bionics*, 3(3):623–639, 2021.
- [Z<sup>+</sup>23] K. Zhang et al. Review of supernumerary robotic limbs. *Journal of Physics: Conference Series*, 2023.
- [Z<sup>+</sup>24] Y. Zhang et al. Motion-compensation control of supernumerary robotic arms subject to human-induced disturbances. *Advanced Intelligent Systems*, 2024.
- [Z<sup>+</sup>26] Y. Zheng et al. Supernumerary robotic limbs: A quantitative analysis and review of design, embodiment, and future challenges. *Journal of Field Robotics*, 2026.
- [ZDA<sup>+</sup>23] Xu Zhao, Wenchao Ding, Yongqi An, Yinglong Du, Tao Yu, Min Li, Ming Tang, and Jinqiao Wang. Fast segment anything. *arXiv preprint arXiv:2306.12156*, 2023.
- [ZGD<sup>+</sup>26] Haoyuan Zhou, Anzhu Gao, Pierre Ducos, Jonathan Eden, and Etienne Burdet. Spatial guidelines for supernumerary robotic limbs in near-body interactions. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, 2026.